Optimal Binding Affinity for Sieving Separation of Propylene from Propane in an Oxyfluoride Anion-Based Metal–Organic Framework

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ABSTRACT:

Highly efficient adsorptive separation of propylene from propane offers an ideal alternative method to replace the energy-intensive cryogenic distillation technology. Molecular sieving-type separation via high-performance adsorbents is targeted for superior selectivity, but the limit in adsorption capacity remains a great challenge. Here, we report an oxyfluoride-based ultramicroporous metal–organic framework **UTSA-400**, $[Ni(WO_2F_4)(pyz)_2]$ (pyz = pyrazine), featuring one-dimensional pore channels that can accommodate the propylene molecules with optimal binding affinity while specifically excluding the propane molecules. The exposed oxide/fluoride pairs in **UTSA-400** serve as strong functional sites for strengthened propylene–host interactions, accounting for a significantly enhanced propylene uptake, while the propane molecules are excluded due to the regulated host framework dynamics. The strong propylene binding enables near-saturation of propylene in the pore confinement at ambient conditions, leading to full utilization of pore space and superior packing density. Combined in situ infrared spectroscopy measurements and dispersion-corrected density functional theory calculations clearly unveil the nature of boosted host–guest binding. Direct production of polymer-grade (>99.5%) propylene with remarkable dynamic productivity is demonstrated by column breakthrough experiments. This work presents an example of pore engineering with atomic precision to break the trade-off in adsorptive separation through guest binding optimization.

Introduction:

The global propylene market has grown rapidly in the past decade due to its wide applications in a great variety of downstream industries including automotive and constructions. The production capacity of propylene, the second highest produced chemical feedstock, was 130 million metric tons worldwide in 2019, and the global propylene market was worth over US$ 90 billion annually. (1,2) The importance of propylene in the petrochemical industry manifests in the high demand of the production of a series of chemicals comprising polypropylene, propylene oxide. Current industrial propylene production relies heavily on the thermal/catalytic cracking of hydrocarbons, yielding propane as a major by-product. (3) The high similarity of propylene and propane in terms of physiochemical properties poses a grand challenge to obtain polymer-grade propylene (>99.5%) and intensifies the energy consumption in the widely used cryogenic distillation technique. (4) Thus, millions of energy cost and millions of tons of carbon emission can be saved if replaced by alternative energy-efficient separation solutions. (5)

Adsorptive separation using porous solid is demonstrated as a promising technique with high energy efficiency. Porous materials, ranging from zeolite, (6,7) carbon materials, (8,9) and metal–organic frameworks (MOFs) (10−17) to hydrogen-bonded organic frameworks (HOFs), (18−20) have been extensively investigated as a propylene/propane splitter. The propylene/propane separation mechanism is generally specified to thermodynamic equilibrium separation, gating effect, and kinetic separation. By virtue of rational design and pore engineering, one can precisely functionalize the pore surface, which creates thermodynamic preference toward propylene, enabling propylene-selective molecular recognition over propane in the competitive adsorption process. (21−24) However, the co-adsorption of propane is typically unavoidable and significantly hinders the selectivity and purity of recovered propylene. Diffusivity-based kinetic separation is proven beneficial by enlarging the difference in diffusion rate and, thus, the adsorption selectivity, especially when synergistically coupled with equilibrium selectivity. (25−29) On the other hand, gating effect-driven separation is found to be effective to realize selective propylene adsorption, thanks to the structural flexibility allowing it to capture propylene through “gate-opening”. (30−32) The sorbents can be properly designed to achieve desired responsiveness. (33,34) But such cooperative pore opening could also lead to co-adsorption of undesired gas and, worse, leakage or “slipping-off” in mixture separation processes. (35,36) More research efforts and insights are still needed to understand this phenomenon. After all, in the ideal scenario, the ultimate goal is to realize molecular sieving, namely, a complete exclusion of the undesired molecule by tailor-made pore architecture featuring suitable size and dynamics.

Molecular sieving is one of the most attractive types of adsorptive separation as the adsorption selectivity is intrinsically very high. (37) Although several propylene sieves were reported to exhibit good propylene separation performance, the capacity/selectivity trade-off still impedes the practical application of these propylene sieves. (38−42) Note that besides the high equilibrium selectivity observed in sieving separations, the desorption selectivity and purity are also nontrivial technical problems, which are directly associated with possible kinetic effects. After all, a clear picture of boosting separation capacity and precise control of framework dynamics deciphered by atomic-level functionalization and characterization is still lacking in this field, which is essential for realizing simultaneous high selectivity and high capacity. Here, we propose the rational design of propylene sieves with optimal thermodynamic binding affinity to overcome the capacity limitation of sieving separation.

We leverage reticular chemistry to report a new MOF, $[Ni(WO_2F_4)(pyz)_2]$ (pyz = pyrazine), termed **UTSA-400**, based on the prototypical **NbOFFIVE-1-Ni**. The isoreticular nature allows us to finely tune the pore chemistry via installing the new oxofluorotungstates $WO_2F_4^{2-}$ as pillar ligands (L). This oxyfluoride-type ligand is specifically targeted to control the M-L-M linkage for enlarged linker tilting and restricted local framework dynamics. Molecular electrostatic potential (MEP) analysis reveals a remarkably more negative MEP on the exposed O/F sites of $WO_2F_4^{2-}$ anions in **UTSA-400**, in contrast to those of $NbOF_5^{2-}$ anions in **NbOFFIVE-1-Ni**. Gas sorption experiments show that the propylene adsorption capacity is substantially enhanced by 544 and 69% at 0.1 and 1 bar, respectively, in **UTSA-400** compared to its analogue with a $NbOF_5^{2-}$ pillar. The larger shift of olefin double bond deformation modes and significant perturbation of WO_2 stretching modes observed in in situ IR spectroscopy provide clear evidence for the strengthened propylene binding in **UTSA-400**. Dispersion-corrected density functional theory (DFT-D) calculations confirm that the propylene molecule is well confined in the channel of **UTSA-400** through strong hydrogen bonding interactions with the $WO_2F_4^{2-}$ anions and maximized vdW contacts with pyz linkers. The boosted dynamic separation capacity and direct production of polymer-grade (>99.5%) propylene are demonstrated in column breakthrough experiments. What is more, the great stability of **UTSA-400** showcases its promise as a C_3H_6 splitter for the industrial transformation of refinery-grade propylene (RGP) to polymer-grade propylene (PGP).

Results and Discussion

Structural Analysis

Solvothermal reactions of pyrazine and a pre-made aqueous solution synthesized from $Ni(NO_3)_2{\cdot}6H_2O$, WO_3, and 48% HF afforded purple plate-shaped crystals of $[Ni(WO_2F_4)(pyz)_2]$ (**UTSA-400**) (Figures 1 and S1). Single crystal diffraction study shows that **UTSA-400** crystalizes in the *P*4/*mmm* space group. The Ni(II) ions are octahedrally coordinated by four different nitrogen atoms from different pyrazine linkers (Ni–N 2.090 Å) and two F atoms from different oxofluorotungstate anions (Ni–O/F 2.018 Å). The two-dimensional $Ni(pyz)_2$ sql layers were further pillared by $WO_2F_4^{2-}$ anions to form a 3D framework with the typical *pcu* topology, which is isoreticular to other pyrazine-based SIFSIX-type MOFs (Figure 1a). In the crystal structure of **UTSA-400**, the pyrazine linkers are disordered at two positions, with a tilting angle of 34.85°, which to the best of our knowledge is greater than that of any other SIFSIX-type MOFs (30.85° in **NbOFFIVE-1-Ni**, (38) 18.27° in **SIFSIX-3-Ni**, (43) and 28.75° in **TaOFFIVE-1-Ni** (44)). The greater tilting leads to a contracted pore size of around 3.0 Å in **UTSA-400**, comparable to that of **NbOFFIVE-1-Ni**, as shown in the pore size distribution calculated by PoreBlazer v4.0 (45) (Figure S2). A close inspection of the pore geometry indicates that eight pyz linkers and four pillars encapsulate a pocket-like cavity with a size of 6.7 × 5.5 × 3.7 Å^3, interconnected by the small pore window (Figures 1d and S4). Furthermore, the large electron cloud of the bulkier W center gives rise to a longer M–O/F_{eq} bond length of 1.835–1.840 Å (subscript eq denoted as O/F in the equatorial position) than those in SiF_6^{2-} and TiF_6^{2-} (Si–F_{eq} 1.613 Å and Ti–F_{eq} 1.81 Å). (43,46) The slightly shorter W–O/F_{eq} bond length in **UTSA-400** compared to $NbOF_5^{2-}$ (1.899 Å) and $TaOF_5^{2-}$ (1.921 Å) is attributed to the hybridization of W–F_{eq} and shorter W═O_{eq} bond. As a result, an F···F diagonal distance (3.244 Å, exclusive of van der Waals radii) lies between the shortest one observed in **NbOFFIVE-1-Ni** and those in **SIFSIX-3-Ni**, **TaOFFIVE-1-Ni**, and **TIFSIX-3-Ni**. The slightly contracted crystal structure of **UTSA-400** is reflected by PXRD patterns in comparison with **NbOFFIVE-1-Ni**, which clearly shows the (100) peak at a slightly higher angle, corresponding to a shorter *c*-axis or Ni···Ni distance (7.831 Å) (Figures 1b and S3 and Table S2). The details of crystallographic information are summarized in Table S2.

Due to the disorder of the $WO_2F_4^{2-}$ anion, its detailed bonding geometry information is not accessible from single crystal diffraction studies. A CSD survey of crystal structures containing

$WO_2F_4^{2-}$ (CSD 5.43 Oct 2021 version) revealed that only 3 out of the total 8 structures show ordered $WO_2F_4^{2-}$ conformations (GIHMES, LIDBEI, and UCOXUF) (Table S3). In these structures, a displacement of the tungsten atom toward two oxide ligands is found, while two oxides adopt a cis position with a short W═O bond length (1.694–1.771 Å) and a O–W–O angle in a range of 97.94–101.33°, as a result of dπ–pπ metal oxide orbital interactions. (47) Such distortion divides the bonding geometries of four F ligands into two groups, F_{trans} and F_{cis}, depending on their orientations with respect to the oxides (Figures 1c and S5). The W–F_{trans} bond exhibits an elongated bond length (1.988–2.063 Å) compared with the W–F_{cis} bond (1.864–1.909 Å). The desymmetrized C_{2v} geometry of the $WO_2F_4^{2-}$ anion distinguishes it from those typical O_h-symmetry hexafluoro-type ligands (e.g., SiF_6^{2-} and TiF_6^{2-}) and other C_{4v}-symmetry oxyfluoroligands (e.g., $NbOF_5^{2-}$ and $TaOF_5^{2-}$), which also brings another level of local framework dynamics in the porous scaffold.

To obtain the solid-state anion geometry in **UTSA-400**, a structure model was adapted from the experimental crystal structure for DFT structural optimization with an assignment of one oxide per unit cell, given the average number of oxides in the formula. The optimized geometry of the $WO_2F_4^{2-}$ anion agrees well with the reported ordered $WO_2F_4^{2-}$ anions, with a W═O bond length of 1.751–1.778 Å, W–F_{trans} bond length of 2.018–2.043 Å, and O–W–O/O–W–F_{trans} angle of 101.9 and 170.2°, respectively (Figure 1c). Infrared spectroscopy was conducted to confirm such a configuration (Figure 1e). We observed a couple of distinct bands located between 1000 and 900 cm^{-1} in two structures, which obviously pertain to the vibrations of inorganic pillar anions, $WO_2F_4^{2-}$ and $NbOF_5^{2-}$. Based on the assignment established in the former studies of oxofluorotungstates and oxofluoroniobium complexes, (48,49) the 967 and 920 cm^{-1} bands in **UTSA-400** are attributed to the stretching (ν_s, ν_{as}) of WO_2 group and the 933 cm^{-1} band in **NbOFFIVE-1-Ni** to the stretching of Nb═O. W–F and Nb–F bonds cannot be directly characterized by IR spectroscopy since their vibrations occur below 650 cm^{-1}, falling beyond the detection range of the infrared MCT-A detector. We have also calculated the stretching frequencies of WO_2 bands including asymmetric and symmetric vibrations (ν_s, ν_{as}), based on our optimized structure (see Figure 1c), the separation between which is sensitive to the bonding geometry (Figure S27). ν_s and ν_{as} bands of WO_2 are predicted to be at 952 and 904 cm^{-1}. The 48 cm^{-1} separation matches perfectly with the experimental result (47 cm^{-1}) collected by infrared spectroscopy (Table S4).

The electrostatic potential (ESP) of optimized $WO_2F_4^{2-}$ and $NbOF_5^{2-}$ anions was calculated to investigate the electronegativity of the equatorial O/F sites. In the $WO_2F_4^{2-}$ anion, the most negative ESP is distributed at the two pairs of O and F_{trans} ligands, which agrees well with its well-established dual functionality as a cis/trans director in the solid state (50) (Figures 1c and S6). With one pair coordinated to the nickel ions, the other pair of O and F_{trans} ligands is exposed to the open pore space and accessible to guest molecules. Similarly, the most negative ESP in $NbOF_5^{2-}$ is located at the pair of oxide and F_{trans} ligand. However, only relatively weaker ESP shared by four equatorial F sites is exposed toward the pore surface. We further mapped the ESP onto the Connolly surface of DFT-optimized **UTSA-400**, which clearly reveals the more negative ESP in the region close to the exposed O/F_{trans} on the surface of the one-dimensional pore channel (Figure 1d). These results suggest that the uncoordinated O/F_{trans} pair in the $WO_2F_4^{2-}$ anion is likely to serve as a strong binding site toward guest molecules.

Characterizations

The phase purity of **UTSA-400** was confirmed by powder X-ray diffraction (PXRD) (Figure S7). The thermogravimetric analysis (TGA) revealed that all guest water molecules are lost in **UTSA-400** at around 140 °C, with a weight loss of 7.2%, corresponding to 2.2 water molecules per formula, and that the structural integrity is maintained up to 240 °C (Figure S8). The framework integrity of **UTSA-400** is preserved after activation, as determined by PXRD, which further demonstrates its thermal stability and prompted us to examine its gas adsorption properties (Figure S7).

Gas Sorption Analysis

The Brunauer–Emmett–Teller surface area of activated **UTSA-400** is calculated as 226 $m^2\ g^{-1}$ from the CO_2 isotherm collected at 298 K (Figures S9 and 10). The surface area of **UTSA-400** is comparable to its isostructural counterparts, such as **NbOFFIVE-1-Ni** (280 $m^2\ g^{-1}$) (38) and **TIFSIX-3-Ni** (218 $m^2\ g^{-1}$). (46) The experimental total pore volume is 0.084 $cm^3\ g^{-1}$, slightly lower than the theoretical value (0.10 $cm^3\ g^{-1}$) calculated from the crystal structure due to marginally insufficient filling of CO_2 at the bottleneck like pore window of the pore system.

Previous investigations of the propylene/propane separation in this series of anion pillared MOFs showed that only **NbOFFIVE-1-Ni** exhibited the molecular sieving effect, thanks to the large tilting of pyz linkers. (38,51) Considering that the tilting angle in **UTSA-400** is even larger, we anticipated that such a sieving phenomenon would be conserved or even enhanced in **UTSA-400**. To validate the relationship between ligand tilting and gating energy threshold, we optimized the open phase (tilting angle $\theta = 0°$) structure models for two MOFs so as to calculate the energy difference between the as-synthesized form and the open phase using first-principles DFT calculations (Figure S11). As expected, the energy barrier of rotating all pyz linkers in as-synthesized **UTSA-400** into open phase is 2.28 eV, which is significantly higher than that of **NbOFFIVE-1-Ni** (1.78 eV), implying the regulated host framework dynamics induced by local linker rotation and the potential propane exclusion in **UTSA-400**.

Motivated by these structural characteristics, we measured the low-pressure C_3H_6 and C_3H_8 gas adsorption isotherms of **UTSA-400** at 298 K (Figure 2a). **UTSA-400** exhibited a large volumetric C_3H_6 uptake of 46.3 and 92.1 $cm^3\ cm^{-3}$ at 0.1 and 1 bar, respectively. In contrast, the adsorption uptake of C_3H_8 at the same condition (2.5 $cm^3\ cm^{-3}$) is almost negligible, which is consistent in 3 consecutive cycles (Figure S13). In comparison, the volumetric C_3H_6 uptakes of **NbOFFIVE-1-Ni** at 0.1 and 1 bar are 8.5 and 54.3 $cm^3\ cm^{-3}$, respectively. In other words, by substituting the $NbOF_5^{2-}$ pillar into the $WO_2F_4^{2-}$ anion, the C_3H_6 uptake capacity is significantly boosted by 544 and 69% at 0.1 and 1 bar, respectively, without sacrificing the feature of excluding propane molecules (Figure 2b). Furthermore, cycling C_3H_6 adsorption experiments for **UTSA-400** revealed that the C_3H_6 uptake was maintained in 5 cycles (Figure 2c). The C_3H_6 uptake of **UTSA-400** and **NbOFFIVE-1-Ni** at 1 bar corresponded to 0.95 and 0.57 molecules per pocket/formula, respectively, suggesting that the pore system of **UTSA-400** is almost saturated by C_3H_6 under ambient conditions (Figures 2e and S14). Based on the adsorption kinetic profile, two

MOFs exhibited comparable diffusion coefficients, suggesting that the restricted framework dynamics in **UTSA-400** has minimal effect on adsorption kinetics (Figure S15).

As a result, based on the determined pore volume, the C_3H_6 packing density in the pore channel of **UTSA-400** is calculated as 0.93 g cm^{-3}, higher than those in other sieving-type materials and even the liquid C_3H_6 density (0.61 g cm^{-3}). The significantly boosted volumetric C_3H_6 uptake of **UTSA-400** is higher than those of widely used zeolites and most reported C_3H_6/C_3H_8 sieving-type materials (zeolite 13X (52) 82.1 cm^3 cm^{-3}, zeolite 4A (52) 73.2 cm^3 cm^{-3}, Si-CHA (53) 90.3 cm^3 cm^{-3}, ITQ-3 (53) 51.9 cm^3 cm^{-3}, Y-dbai (42) 75.9 cm^3 cm^{-3}, Y-abtc (39) 64.6 cm^3 cm^{-3}, and Co-gallate (40) 66.6 cm^3 cm^{-3}) and almost comparable to the benchmark HIAM-301 (41) (93.0 cm^3 cm^{-3}) (Figure 2d). Such unprecedented high loading and packing density unveil the strong binding nature with C_3H_6 in **UTSA-400**.

Ideal adsorbed solution selectivity (IAST) theory was then employed to evaluate the adsorption selectivity toward equimolar C_3H_6/C_3H_8 mixture at 1 bar and 298 K (Figures S16 and 17). The calculated selectivity of over 10^7 at ambient condition exceeded that of all other reported MOFs, including those showing molecular sieving effect, such as Co-gallate (40) (333) and HIAM-301 (41) (150) (Table S6). Note that for adsorbents showing sieving separation performance, the calculated IAST selectivity is only for qualitative comparison as large error could be generated from the low uptake of the unadsorbed molecule. Nonetheless, the ultrahigh selectivity demonstrated its great potential of **UTSA-400** for efficient C_3H_6/C_3H_8 sieving separation.

To validate the binding affinity toward C_3H_6 in **UTSA-400**, the isosteric heat of adsorption (Q_{st}) of **UTSA-400** for C_3H_6 was calculated by the virial fitting method. The calculated Q_{st} for C_3H_6 is 60.5 kJ mol^{-1} at near-zero coverage (Figures S18–20). In contrast, the lower Q_{st} for C_3H_6 in **NbOFFIVE-1-Ni** (52 kJ mol^{-1}) quantitively confirmed the optimized binding of C_3H_6 in **UTSA-400**. The Q_{st} of **UTSA-400** for C_3H_6 is, to the best of our knowledge, the highest in the molecular sieving-type MOFs and even higher than or comparable to those of zeolites and some MOFs with open metal sites, such as zeolite 13X (52) (42.5 kJ mol^{-1}), zeolite 4A (52) (29.9 kJ mol^{-1}), Ni-NP (24) (57 kJ mol^{-1}), Fe_2(dobdc) (54) (44 kJ mol^{-1}), Co_2(*m*-dobdc) (22) (53.1 kJ mol^{-1}), and Ni_2(*m*-dobdc) (22) (55.1 kJ mol^{-1}) (Figure 2d, Table S6). Such strong adsorption enthalpy from purely intermolecular noncovalent interactions without π-complexation effect showcases the potential of pore chemistry engineering to realize optimal binding affinity for simultaneous high uptake capacity and high selectivity.

In Situ Infrared Spectroscopy Study

We carried out in situ infrared spectroscopy to further probe the mechanistic insights into the adsorption of propylene in **UTSA-400** as well as **NbOFFIVE-1-Ni** for comparison. The sample was first activated by heating up to 150 °C under vacuum and then cooled back to 24 °C for recording the reference spectrum and subsequently loading propylene. The IR spectra of activated **UTSA-400** and **NbOFFIVE-1-Ni** above 700 cm^{-1} (see bottom lines of Figures 3 and S24) are dominated by the similar characteristic bands that are associated with the vibrations of their same organic linker pyrazine including ν(CH) at 3100–2900 cm^{-1}, ν(CC)/δ(CH)$_{ip}$ at 1450–1350 cm^{-1}, δ(CH)$_{ip}$/ν(CN)/δ(ring) at 1200–1050 cm^{-1}, and δ(CH)$_{oop}$ at ∼830 cm^{-1}. (55,56) The infrared bands of the pyz linker in MOFs have been studied in our prior work, (55) and detailed assignments are summarized in Table S4. The samples were first exposed to propylene at ∼760 torr, followed by evacuation under vacuum (see Figure 3). The adsorbed propylene is typified by their characteristic bands including ν(CH, CH_2, and CH_3) at 3200–2900 cm^{-1}, (57) ν(C═C) at 1641 cm^{-1}, (58,59) $\delta_{as,s}(CH_3)$ at 1460–1350 cm^{-1}, (57) and σ(CH═CH_2) and ω(CH═CH_2) at 1000–880 cm^{-1} (see calculated vibrations of propylene molecules in Figure S26 and Table S5). It is clear that more propylene is adsorbed in **UTSA-400** as seen by the higher intensity of its vibrational bands. Note that ν(CH, CH_2, and CH_3) bands of adsorbed propylene at 3200–2800 cm^{-1} were obscured by the strongly perturbed pyrazine CH stretching bands that occur in the same region. Thus, we focus our analysis on the region below 1800 cm^{-1}, where there is less overlap between host and guest absorption bands. The band at 1641 cm^{-1} is due to the C═C stretching of propylene, the frequency of which has been shown to be sensitive to the binding environment of the olefin double bond upon interaction with the adsorbent. (58,59) The occurrence of the ν(C═C) band at the same position 1641 cm^{-1} (−12 cm^{-1} lower than that of the free propylene molecule at 1653 cm^{-1}) in both **UTSA-400** and **NbOFFIVE-1-Ni** suggests that the olefin double bond interacts with the same chemical group in two structures, which can be mainly ascribed to the pyz linker. (57,58) The strong perturbations of pyz linker bands $\nu(CH)_{ph}$, $\delta(CH)_{ip}$/ν(CN)/δ(ring), and $\delta(CH)_{oop}$ were observed in the difference spectra of Figure 3 as indicated by the derivative-like features, providing the evidence for such interactions. As propylene departs from the frameworks upon evacuation of the gas phase, the perturbations gradually disappear, confirming that propylene is reversibly adsorbed. We further noticed that the wagging band of the hydrocarbon group on the olefin double bond, namely, ω(CH═CH_2), occurs at different frequencies in **UTSA-400** and **NbOFFIVE-1-Ni** (886 and 892 cm^{-1}, shown in Figures 3 and S25), indicating that the interacting environment of CH or CH_2 groups of propylene differs in two materials, which is evidently related to WO_2F_4 and $NbOF_5$ moieties. ω(CH═CH_2) in both materials exhibits a downward shift with respect to the gas phase value at 912 cm^{-1}. The larger shift in **UTSA-400** implies a stronger attractive interaction between CH/CH_2 and WO_2F_4 that decreases the wagging deformation potential and thus reduces its vibrational frequency. (60) The observation of $\nu_{s,as}(WO_2)$ bands at 1000–900 cm^{-1} being strongly perturbed offers the proof for the interaction of propylene with the exposed oxide of the $WO_2F_4^{2-}$ anion, as verified by our following modeling studies. In contrast, the ν(NbO) band in $NbOF_5^{2-}$ of **NbOFFIVE-1-Ni** is not affected since the oxide only serves as the bridging ligand connected to both Ni^{2+} and Nb^{5+}. In addition, we also noticed a small frequency difference (∼2 cm^{-1}, see Table S5) in one of CH_3 deformation bands, which suggests that the CH_3 binding environment could be slightly different.

We further conducted first-principles dispersion-corrected DFT (DFT-D) calculations to structurally investigate the binding orientations of C_3H_6 and elucidate involved intermolecular interactions in **UTSA-400**. Given the statistic average amount based on the chemical formula one oxide per Ni/pocket, initial coordinates featuring one exposed oxide in each pocket unit were adopted for subsequent calculations. The C_3H_6 molecule is found to be confined appropriately in the gourd-like channel via multiple noncovalent intermolecular interactions with pyz linkers and the $WO_2F_4^{2-}$ anions (Figure 4a). A close inspection of the orientation of C_3H_6 molecule reveals that the smallest cross section of C_3H_6 (the intersection between the methyl group and vinyl group) is located at the bottleneck window (surrounded by the equatorial O/F sites) (Figure 4b). Such binding configuration not only reduces steric hinderance but also maximizes close contacts with the equatorial O/F sites and pyz ligands. Specifically, strong hydrogen bonding interactions are observed with oxide/F_{trans} ($d_{H\cdots O/F}$ = 2.215–2.831 Å, C–H···O/F angle of 110.8–144.5°) and F_{cis} ($d_{H\cdots F}$ = 2.216–2.469 Å, C–H···F angle of 95.4–134.5°) (Figure 4b,c). What is more, the vinyl group of propylene also forms multiple weak C–H·· π van der Waals interactions with the pyz linkers ($d_{H\cdots C}$ = 2.884–3.330 Å), which rationalizes the large shift of its deformation band. Very similar binding energies were found in three different propylene binding orientations with respect to the position of oxide in the obtained **UTSA-400** (Figure S21). A similar binding orientation is also found for C_3H_6 in **NbOFFIVE-1-Ni** but with relatively elongated hydrogen bonding distances ($d_{H\cdots F}$ = 2.316–2.710 Å), implying weaker binding affinity (Figure S22a,b). The calculated C_3H_6 binding orientation agrees well with the previously determined position of C_3H_6 in **NbOFFIVE-1-Ni** by single crystal XRD (61) (Figure S22c). The derived static binding energies for C_3H_6 in **UTSA-400** and **NbOFFIVE-1-Ni** were then calculated as 68.7 and 54.6 kJ mol^{-1}, respectively, which unambiguously demonstrated the boosted binding energy in **UTSA-400**, thanks to the strengthened H-bonding and maximized vdW contacts.

Hirshfeld surface analysis was employed to examine the existence of multiple noncovalent interactions by mapping d_{norm} onto the Hirshfeld surface of the confined C_3H_6 using Crystal Explorer 21 (62) (Figure 4c). The quantity of d_{norm} is computed as the sum of normalized d_i (distance from a point on the Hirschfeld surface to the nearest nuclei inside the surface) and d_e (the distance to the nearest external nucleus) by the van der Waals radii for the involved atoms. A negative value of d_{norm} indicates that the contact distance is shorter than the sum of vdW radii, which is reflected in red to show the strong intermolecular contacts (Figure 4c). The calculated minimum value of d_{norm} in two MOFs (−0.2871 for **UTSA-400**, −0.1831 for **NbOFFIVE-1-Ni**) suggests the stronger binding affinity for C_3H_6 in **UTSA-400**. Next, the host–guest interaction components in both propylene-loaded structures were analyzed by the derived 2D fingerprint plots (Figure S23). Three bright blue spikes that correspond to H···O, H···F, and H···H contacts are observed in **UTSA-400**, which are attributed to the contribution from C–H···O (7.6%), C–H···F (23.8%), and C–H···π (42.0%) interactions. As a comparison, the absence of H···O contacts and less strong C–H···O/F contribution can be deduced from the 2D fingerprint plot of **NbOFFIVE-1-Ni**.

Dynamic Breakthrough Experiments

To evaluate the dynamic separation performance, fixed-bed breakthrough experiments were carried out for the propylene/propane (50:50 v/v) mixture in **UTSA-400** and **NbOFFIVE-1-Ni**. An equimolar C_3H_6/C_3H_8 mixture was flowed over a packed column of the activated solid with a rate of 2 mL min^{-1} at 298 K. A complete separation of propane from propylene was observed for both materials since the propane molecules elute from the column directly, while propylene is selectively captured with no detectable amount in the outlet until saturation (Figure 5a). The breakthrough time of C_3H_6 in the column packed with **UTSA-400** is around 18 min g^{-1}, giving a dynamic propylene productivity of 46.7 L/L sorbents, higher than that of **NbOFFIVE-1-Ni** (30.7 L/L sorbents). The dynamic C_3H_6 productivity of **UTSA-400** is also higher than those of other sieving-type MOFs such as Co-gallate (40) (36.4 v/v) and Y-abtc (39) (41.1 v/v). What is more, both materials can be easily regenerated by purging helium (40 mL min^{-1}) at RT for less than 30 min, and very low propane component was found in the desorption profiles of both materials

(Figures 5b and S28). The purity of desorbed propylene in **NbOFFIVE-1-Ni** and **UTSA-400** is calculated to be as high as 98.7 and 99.7%, respectively. Thus, the direct production of PGP (>99.5%) in one adsorption–desorption cycle is enabled by **UTSA-400** under ambient conditions. Three different methods have been adopted to regenerate **UTSA-400** in cycling separations: helium purge, typical temperature swing adsorption (TSA) process (heating at 100 °C), and combined vacuum and TSA process, which all show retained performance (Figures S29–31). The feasible regeneration indicates the promise of the implementation in practical separation processes. Note that although the regeneration energy is reported to be directly associated with the heat of adsorption, in the technical regeneration process, the thermal energy for heating the bed is also a contributor of the total energy. As estimated by Berger et al., (63) at a given desorption temperature, a minimum energy for a TSA cycle can be found at a certain heat of adsorption. Future research might also investigate the correlation of heat of adsorption, heat capacity of MOFs, and regeneration energy to find the optimal Q_{st} for a good balance of binding affinity and regeneration cost.

Stability

To evaluate the stability of **UTSA-400** in practical conditions, the structural integrity of **UTSA-400** after exposure to water liquid and vapor was examined by PXRD and gas sorption experiments. The PXRD patterns show that the structure is stable after water immersion, water vapor treatment, and exposure in air (Figure S32). Over 95% of propylene uptakes at ambient conditions is preserved, indicating good water stability (Figure S34). Moreover, **UTSA-400** is also chemically stable in the range of pH = 1–12 for 1 day, as shown by PXRD patterns. 91 and 93% of propylene uptake are retained after being treated by pH = 1 and 12 for 1 day (Figures S33 and 34). The good hydrolytic and chemical stability underscore the great potential of **UTSA-400** in practical separation processes.

Conclusions

In summary, we successfully demonstrated a new approach to obtain optimal binding affinity for highly efficient molecular sieving separation of propylene from propane in **UTSA-400**, based on a new oxyfluoride anion, $WO_2F_4^{2-}$. The quest for the molecular sieving C_3H_6/C_3H_8 separation with high C_3H_6 uptake and diminished impurity co-adsorption requires extremely precise engineering of pore size, host dynamics, and functionality. In this context, by virtue of reticular chemistry, a feasible pore engineering method is discovered to optimize the host–guest binding interactions and achieve high propylene adsorption capacity so as to overcome the trade-off and capacity limit of the current molecular sieves. The precise recognition of and response toward the subtle difference of propylene and propane are attributed to the optimized dynamic behavior of host framework. This work will facilitate the rational design of the next-generation porous materials to address the challenges in chemical separations in an energy-efficient way.

Supporting Information

Experimental procedures, additional structural parameters and figures, gas adsorption data, PXRD patterns, TGA curves, SEM images, adsorption kinetic profile, IR spectra, and additional tables.

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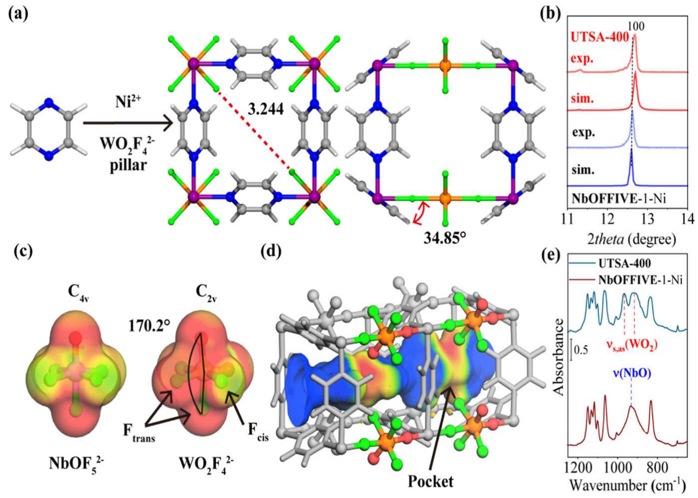

Figure 1. (a) Crystal structure of **UTSA-400** viewed along the *c*- and *b*- axis. Ni, W, C, N, O, F, and H are represented by purple, orange, gray, red, green, and white, respectively. (b) PXRD patterns of **UTSA-400** and **NbOFFIVE-1-Ni** in a lower angle region. (c) Optimized structure and molecular electrostatic potential (MEP) of $NbOF_5^-$ and $WO_2F_4^{2-}$ anions. (d) Connolly surface of **UTSA-400** mapped with electrostatic potential with a probe of 1.2 Å. Each pocket is interconnected by the window surrounded by four pyz linkers. The negative and positive potential are shown in red and blue. (e) Partial IR spectra of activated **UTSA-400** (top) and **NbOFFIVE-1-Ni** (bottom) showing the vibrational signals of the anion pillars. For full spectra and assignment, see Figure 4 and Supporting Information. Notation and acronym: ν = stretching, as = asymmetric, and s = symmetric.

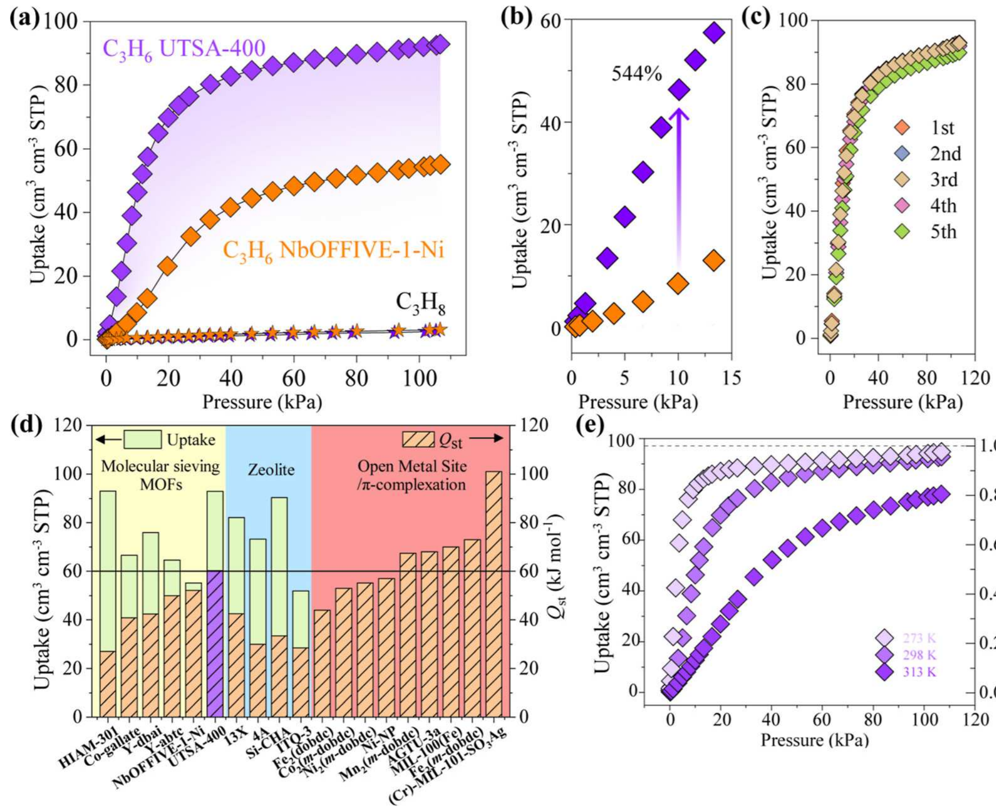

Figure 2. (a) Single-component adsorption isotherms and (b) low-pressure range of C_3H_6 and C_3H_8 for **UTSA-400** and **NbOFFIVE-1-Ni** at 298 K. (c) Five consecutive cycles of C_3H_6 adsorption measurements in **UTSA-400**. (d) Comparison of C_3H_6 uptake capacities and Q_{st} among molecular-sieving type MOFs, zeolites, and MOFs with open metal sites/π-complexation. Note that the isotherm measurement for zeolite is at 303 K. (e) Single-component adsorption isotherms of C_3H_6 at 273–313 K for **UTSA-400** with corresponding adsorption amount per pocket indicated.

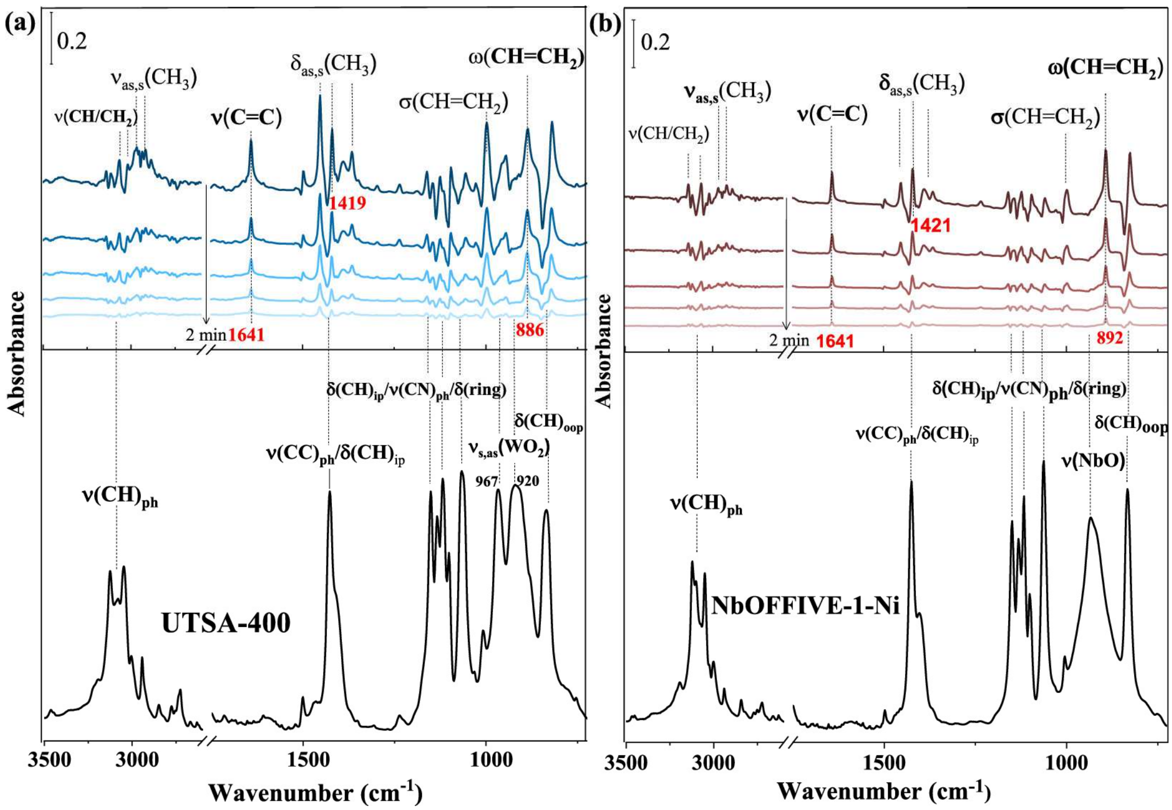

Figure 3. IR spectra showing the adsorbed propylene in (a) **UTSA-400** and (b) **NbOFFIVE-1-Ni** loaded at ~760 torr for ~10 min. Top panels: difference spectra of unloading propylene from two MOFs within ~2 min upon evacuation of the gas phase under vacuum; each spectrum is referenced to activated MOFs. The gas phase signal of propylene at ~760 torr in the initial spectra (prior to evacuation) was subtracted. Bottom panels: IR spectra of activated (empty) **UTSA-400** and **NbOFFIVE-1-Ni**, referenced to the pure KBr pellet under vacuum. Notation and acronym: ν = stretching, δ = deformation, σ = twisting, ω = wagging, ip = in plane, oop = out of plane, as = asymmetric, s = symmetric, and ph = phenyl.

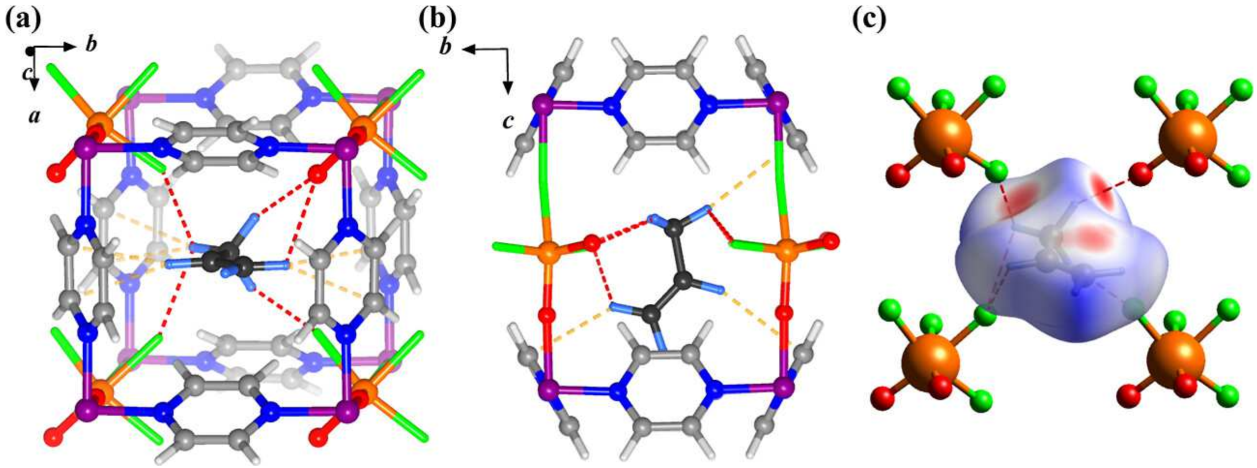

Figure 4. Perspective view (a) and top view (b) of the DFT-D calculated binding site of C_3H_6 in **UTSA-400** with hydrogen bonding interactions and van der Waals interactions shown in red and yellow, respectively. (c) Hirshfeld surface analysis mapped with d_{norm}, highlighting strong intermolecular contacts in red (contact distance shorter than the sum of vdW radii).

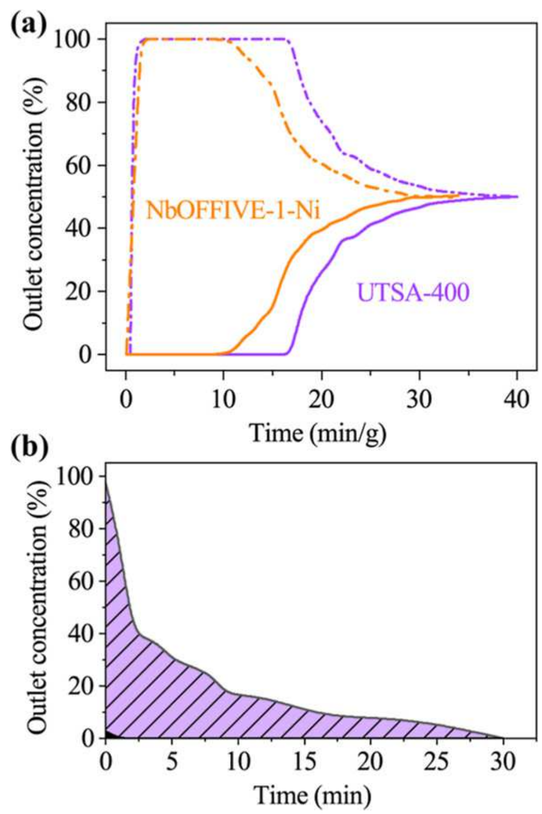

Figure 5. (a) Breakthrough curves for **NbOFFIVE-1-Ni** (orange) and **UTSA-400** (purple) for an equimolar binary mixture of C_3H_6 (solid line)/C_3H_8 (dash line) at 298 K and 1 bar. (b) Concentration curve of the desorbed C_3H_6 (purple) from **UTSA-400** during the regeneration process.

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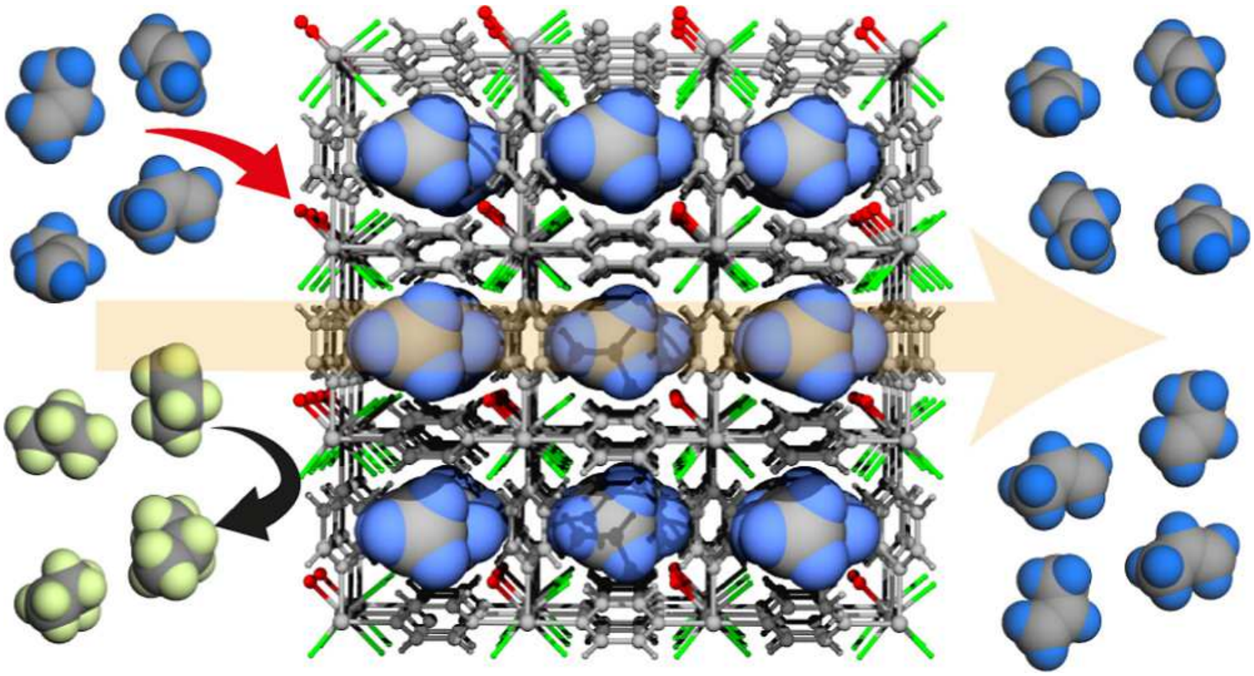

Supporting Information

Optimal Binding Affinity for Sieving Separation of Propylene from Propane in an Oxyfluoride Anion-Based Metal-Organic Framework

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Materials and General Methods.

All reagents were commercially available and used without further purification. Thermogravimetric analysis was performed using a TA-Q50 system. The **NbOFFIVE-1-Ni** sample was synthesized according to literature methods.[1] Powder X-ray diffraction (PXRD) patterns were recorded on a PANalytical X'Pert[3] powder diffractometer equipped with a Cu sealed tube (λ = 1.54184 Å) at 40 kV and 40 mA over the 2θ range of 5–40°.

Synthesis of UTSA-400: $Ni(NO_3)_2 \cdot 6H_2O$ (0.0875g, 0.3 mmol) and WO_3 (0.070g, 0.3 mmol) were dispersed in a mixture containing 1 mL of water and 0.2 mL of HF (48%, aqueous). The mixture was kept into a Parr Teflon-lined reaction vessel at 160 ^oC for 48 hours to ensure all WO_3 were reacted, which was allowed to cool to RT, and the yielded green solution should be homogeneously clear. To the solution, an aqueous solution of pyrazine (0.200g, 2.5 mmol) in 1 mL of water was added and the mixture was sealed into a Parr Teflon-lined reaction vessel at 140 ^oC for 24 hours. (Yield: 90%, based on W) The obtained purple crystals were washed with water several times, followed by solvent exchange in methanol or acetone. Caution: HF is toxic and corrosive. It must be handled with extreme caution and the appropriate protective gear.

X-ray crystallography. The single crystals of as-synthesized **UTSA-400** suitable for X-ray diffraction were mounted in Paratone oil onto a glass fiber and frozen under a nitrogen cold stream. The data were collected at 100 K using a Rigaku AFC12/Saturn 724 CCD fitted with Cu Kα radiation (λ = 1.54184 Å). Data collection and unit cell refinement were executed by using Crystal Clear software. Data processing and absorption correction, giving minimum and maximum transmission factors, were accomplished with CrysAlisPro and SCALE3 ABSPACK, respectively. The structure was solved by direct methods and refined on F2 using full-matrix, least-squares techniques with SHELXL. All non-hydrogen atoms were refined with anisotropic displacement parameters. All hydrogen atom positions were determined by geometry and refined by a riding model. Due to the positional disorder of O/F in $WO_2F_4^{2}$ anion, the axial O2 and F2 atoms of $WO_2F_4^{2}$ anion are refined at the same position and with the same thermal parameters (EXYZ and EADP). The equatorial O1/F1, O3/F3 atoms of $WO_2F_4^{2}$ anion are refined at the same position and with the same thermal parameters (EXYZ and EADP). The occupancy ratio of O and F at the same sites at the equatorial positions is set to 1:3 given that there are one oxide and three fluorides based on the chemical formula.

Crystallographic data and refinement information are summarized in Table S1. CCDC 2212852 contains the supplementary crystallographic data of **UTSA-400**. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam. ac.uk/data_request/cif.

Gas Sorption Measurement: The adsorption isotherms were measured with an automatic volumetric adsorption apparatus ASAP 2020. The acetone exchanged samples (about 200 mg) were placed in the sample tube and activated overnight at 100 °C. The equilibrium time should be at least 30 seconds to guarantee equilibrium is reached.

***in situ* Infrared (IR) spectroscopy**: *In situ* IR measurements were performed on a Nicolet 6700 FTIR spectrometer using a liquid N_2-cooled mercury cadmium telluride (MCT-A) detector. The spectrometer is equipped with a vacuum cell that is placed in the main compartment with the sample at the focal point of the infrared beam. The samples (~5 mg) were gently pressed onto KBr pellet and placed into a cell that is connected to gas feeding line and vacuum line for evacuation. The samples were activated upon evacuation under vacuum at 150 ºC for >3 h, and then cooled back to room temperature (24 °C) for propene adsorption measurement.

Breakthrough experiments: These experiments were carried out in a dynamic gas breakthrough set-up. A stainless-steel column with inner dimensions of 9 mm and a length of 150 mm was used for sample packing. Microcrystalline samples (0.85 g for **UTSA-400**, 0.75g for **NbOFFIVE**-1-Ni) were air dried and then packed into the columns. The columns were activated in vacuum oven at 105°C. The mixed gas flow and pressure were controlled by using a pressure controller valve and a mass flow controller. Outlet effluent from the column was continuously monitored using gas chromatography (GC-2014, Shimadzu) with a thermal conductivity detector. The column packed with activated sample was first purged with helium gas flow (50 mL min^{-1}) for 1 h at room temperature. The mixed gas flow rate during the breakthrough process is 2 mL min^{-1} at 1 bar. After the breakthrough experiment, the sample was purged with helium gas flow (40 mL min^{-1}) at 298 K, from which the outlet effluent was monitored. The column can be regenerated with or without vacuum at 373 K for 2 h, and then purged with helium gas flow (50 mL min^{-1}) at 298 K for 20 min before next cycle.

Computational details: All the calculations were performed in first-principle density functional theory (DFT) in the CASTEP code in the Material Studio 2019 package.[2] (BIOVIA, Dassault Systèmes, Materials Studio 2019, Dassault Systems, San Diego, 2018.) The calculation of ESP and mapping on the pore surface are described in our previous report.[3] The IR simulations for propylene and WO2F4 anion were performed by placing the molecule in a simulation box with a size of 20×20×20 Å^3, which is large enough to isolate the molecule and avoid any long-range intermolecular interactions. The generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE)[4] functional and on-the-fly generated ultrasoft pseudopotentials[5] were used. Grimme (G06) semiempirical methods to describe the long-range van der Waals interactions.[6] Considering the C6 parameter and van der Waals radii R_0 in DFT-D2 are only defined for elements H-Xe, the parameter of W is obtained from Xe. A cutoff energy of 650 eV and a $2 \times 2 \times 1$ *k*-point mesh were found to be enough for the total energy to converge within 1×10^{-5} eV atom $^{-1}$. Full geometry optimizations were performed on the structures loaded with one C_3H_6 molecules. The static binding energy was calculated: $\Delta E = E(\text{MOF}) + E(\text{gas}) - E(\text{MOF} + \text{gas})$.

Fitting of pure component isotherms

The experimentally collected isotherms for C_3H_6 and C_3H_8 at 298 K were fitted with the dual-site Langmuir-Freundlich equation.

$$q = q_{A,sat}\frac{b_A p^{c_A}}{1 + b_A p^{c_A}} + q_{B,sat}\frac{b_B p^{c_B}}{1 + b_B p^{c_B}}$$

The Langmuir parameters for each site is temperature-dependent,

$$b_A = b_{A0} exp\left(\frac{E_A}{RT}\right); b_B = b_{B0} exp\left(\frac{E_B}{RT}\right)$$

where p (unit: kPa) is the pressure of the bulk gas at equilibrium with the adsorbed phase, q (unit: mmol g^{-1}) is the adsorbed amount per mass of adsorbent, q_{sat} (unit: mmol/g) is the saturation capacities, c (unit: kPa^{-1}) is the affinity coefficient.

Calculation of isosteric heat of adsorption (Q_{st})

The virial equation is employed to fit the adsorption isotherms:

$$\ln P = \ln N + \frac{1}{T}\sum_{i=0}^{m} a_i N^i + \sum_{i=0}^{n} b_i N^i$$

$$Q_{st} = -R\sum_{i=0}^{m} a_i N^i$$

The isosteric heat of adsorption (Q_{st}) is determined based on the fitting parameters:

IAST calculations of adsorption selectivity

For the separation of a binary mixture of components C_3H_6, and C_3H_8, the adsorption selectivity is defined by

$$S_{ads} = \frac{q_1/q_2}{y_1/y_2}$$

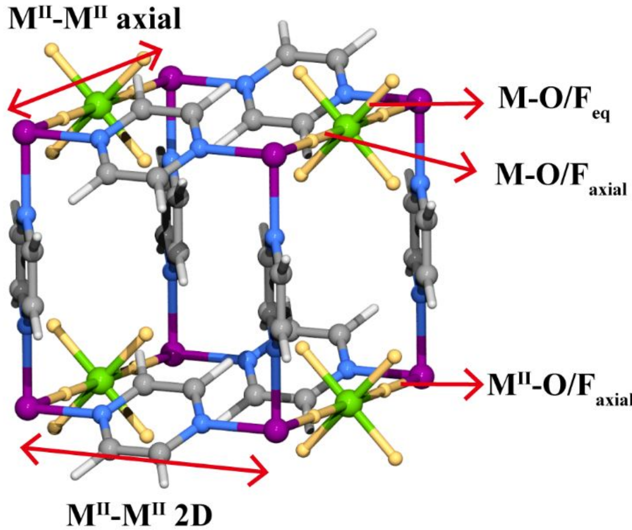
The q_1, and q_2 represent the molar loadings of C_3H_6, and C_3H_8, expressed in mol kg^{-1}, within the MOF that is in equilibrium with a bulk fluid mixture with mole fractions y_1, and $y_2 = 1-y_1$. The molar loadings, also called gravimetric uptake capacities, are usually expressed with the units mol kg^{-1}. The IAST calculations of 50/50 mixture adsorption taking the mole fractions $y_1 = 0.5$ and $y_2 = 1-y_1= 0.5$ for a range of pressures up to 101 kPa and 298 K were performed.

Table S1. Crystal structure data and structure refinements of **UTSA-400**.

Complex	**As synthesized UTSA-400**
CCDC #	2212852
Formula	$C_2H_2FNNi_{0.25}O_{0.5}W_{0.25}$
Formula weight	127.69
Temperature (K)	100
Crystal system	tetragonal
Space group	P4/mmm
a/Å	6.96700(10)
b/Å	6.96700(10)
c/Å	7.8308(2)
α/°	90
β/°	90
γ/°	90
Z	4
F(000)	238.0
D_c/g cm^{-3}	2.231
reflns coll.	1913
unique reflns.	250
R_{int}	0.0326
R_1 [$I > 2\sigma(I)$][a]	0.0269
wR_2 [$I > 2\sigma(I)$][b]	0.0712
R_1 (all data)	0.0269
wR_2 (all data)	0.0712
GOF	1.072
Largest diff. peak/hole / e Å^{-3}	0.60/-1.35

[a] $R_1=\sum||F_o|-|F_c||/\sum|F_o|$. [b] $wR_2=[\sum w(F_o^2-F_c^2)^2/\sum w(F_o^2)^2]^{1/2}$.

Table S2. Detailed structural parameters of nickel pyrazine based-SIFSIX type MOFs.



	UTSA-400	NbOFFIVE-1-Ni	SIFSIX-3-Ni	TIFSIX-3-Ni	TaOFFIVE-1-Ni
CCDC No.	2212852	1447953	1517365	ref[7-8]	2085735
M-O/F_{eq} /Å	1.835-1.840	1.899	1.639	1.81	1.921-1.931
M-O/F_{axial} /Å	1.897	1.941	1.761	N/A	1.944
M^{II}-O/F /Å	2.018	2.000	1.988	N/A	1.995
M^{II}-M^{II} 2D /Å	6.967	7.030	7.001	N/A	7.020
M^{II}-M^{II} axial /Å	7.831	7.882	7.498	7.65	7.878
F-F distance /Å	3.244	3.210	3.683	3.61	3.266
Pyz tilt angle /°	34.85	30.53	18.27	N/A	28.75

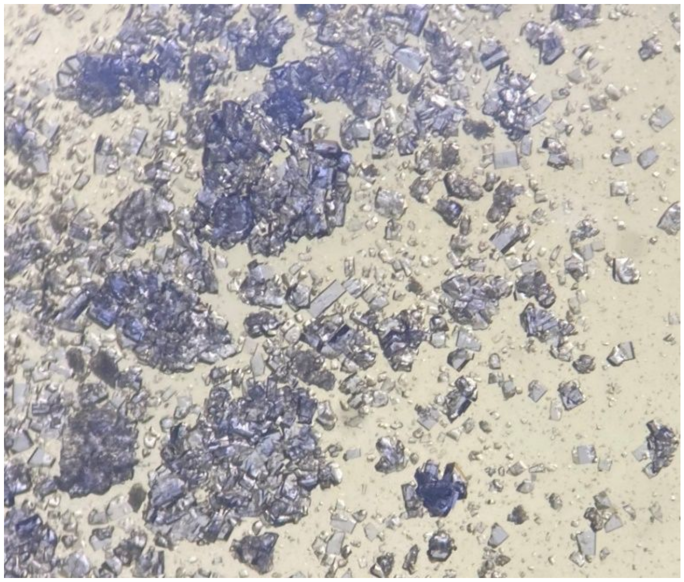


Figure S1. Optical image of as-synthesized **UTSA-400** crystals.

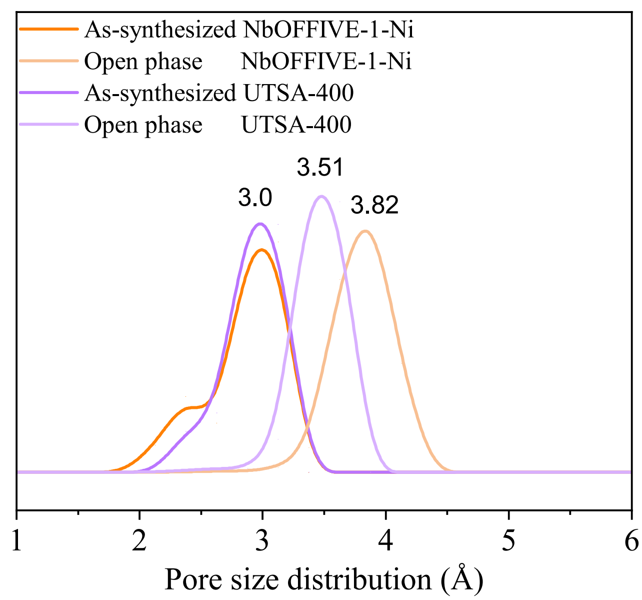


Figure S2. Calculated pore size distribution (PSD). The as-synthesized crystal structures were adapted with guest molecules removed. The PSD of optimized open phase **UTSA-400** and **NbOFFIVE-1-Ni** were also calculated for better comparison. The contracted pore size exhibited in open phase **UTSA-400** indicated a slightly smaller pore channel attributed to the distorted $WO_2F_4^{2-}$ geometry and elongated W-F_{trans} bonds.

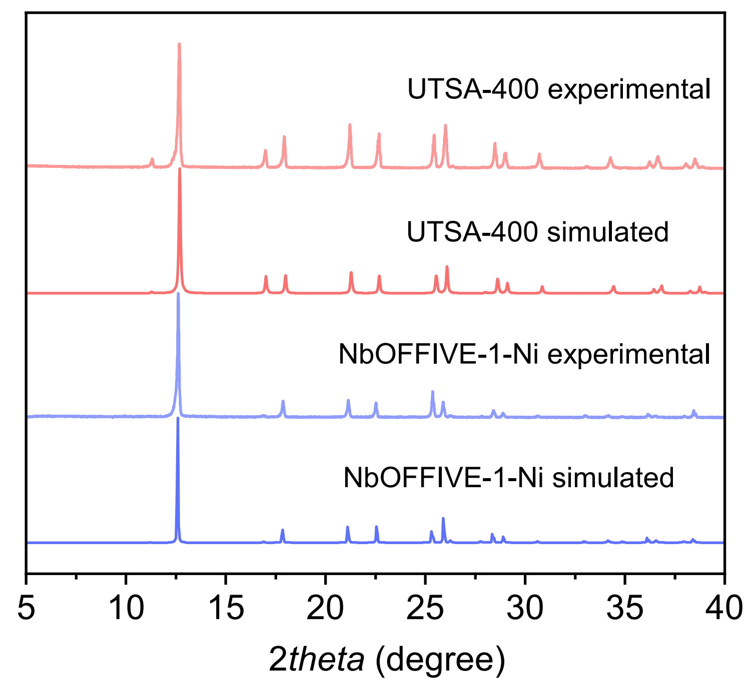


Figure S3. PXRD patterns of simulated and as-synthesized **UTSA-400** and **NbOFFIVE-1-Ni**.

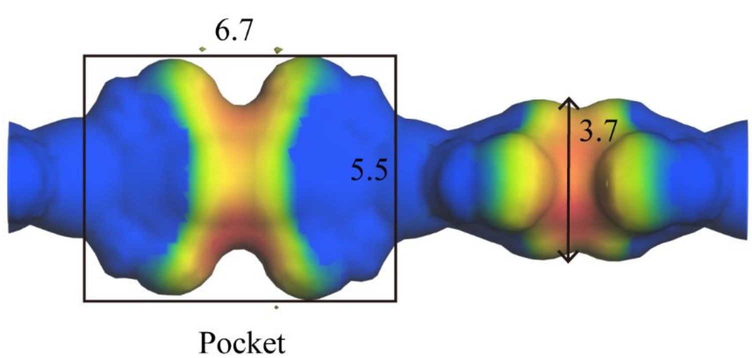


Figure S4. Pore geometry and dimensions of **UTSA-400** (unit in Å) with a probe of 1.2 Å .

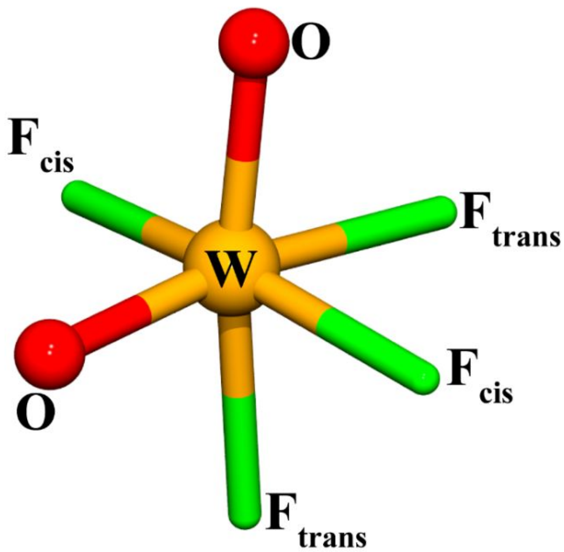


Figure S5. Geometry of $WO_2F_4^{2-}$ anion.

Table S3. Summary of ordered $WO_2F_4^{2-}$ anion reported.

CCDC Refcode	**GIHMES**	**LIDBEI**	**UCOXUF**	**DFT optimized UTSA-400**
W-O length/Å	1.731-1.746	W1 1.701-1.761	1.714-1.771	1.751-1.778
		W2 1.694-1.730		
W-F_{cis} length/Å	1.900-1.902	W1 1.864	1.903-1.909	1.946
		W2 1.873		
W-F_{trans} length/Å	2.017-2.019	W1 2.002-2.045	2.029-2.059	2.043 2.018
		W2 1.988-2.063		
O-W-F_{tran} angle/°	168.90-170.07	W1 170.27-166.58	167.83-170.26	170.2 166.0
		W2 166.89 173.17		
F_{cis}-W-F_{cis} angle/°	162.42	W1 162.68	162.99	160.5
		W2 160.34		
O-W-O angle/°	100.87	W1 101.33	100.71	101.98
		W2 97.94		

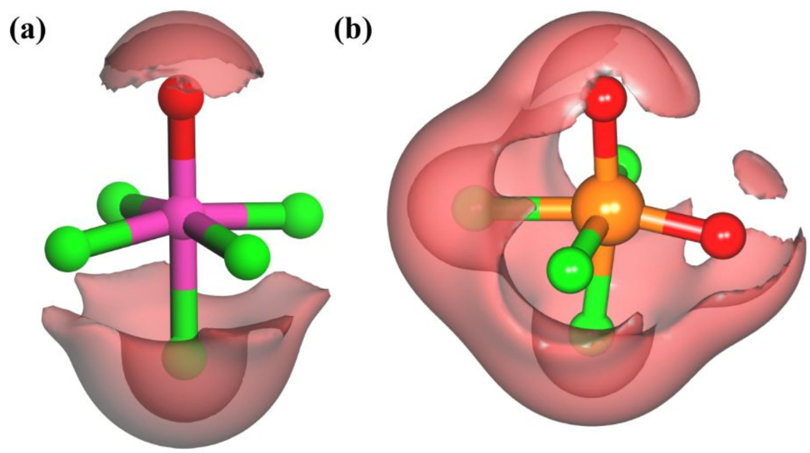


Figure S6. Isosurface maps of the MEP for (a) $NbOF_5^{2-}$ and (b) $WO_2F_4^{2-}$ (isovalue = -0.38 au) anions . Red colors represent the negative region of MEP.

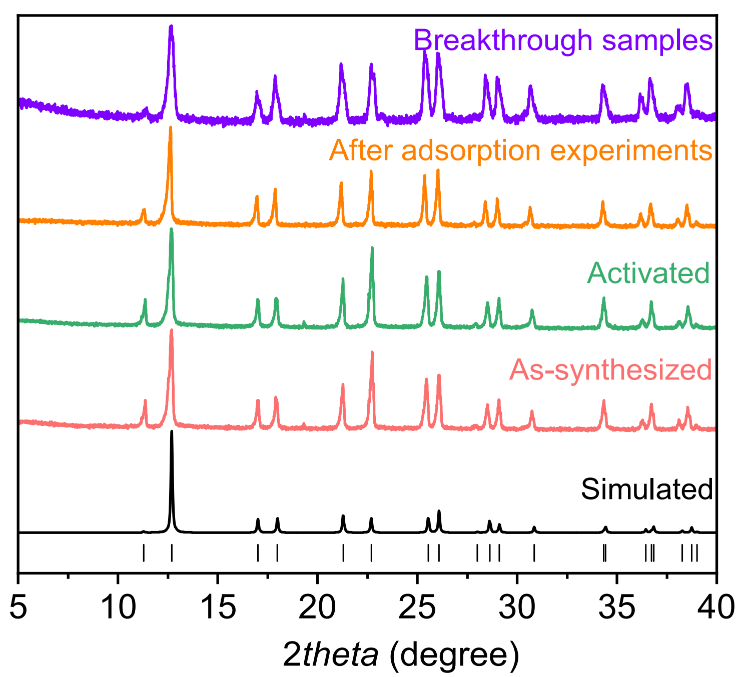


Figure S7. PXRD patterns of **UTSA-400**.

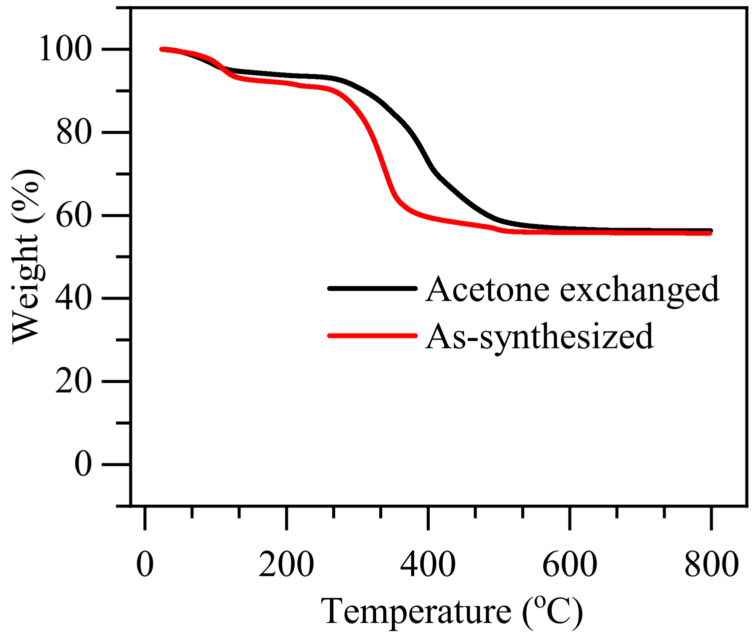


Figure S8. TGA curve of as-synthesized and acetone-exchanged **UTSA-400**.

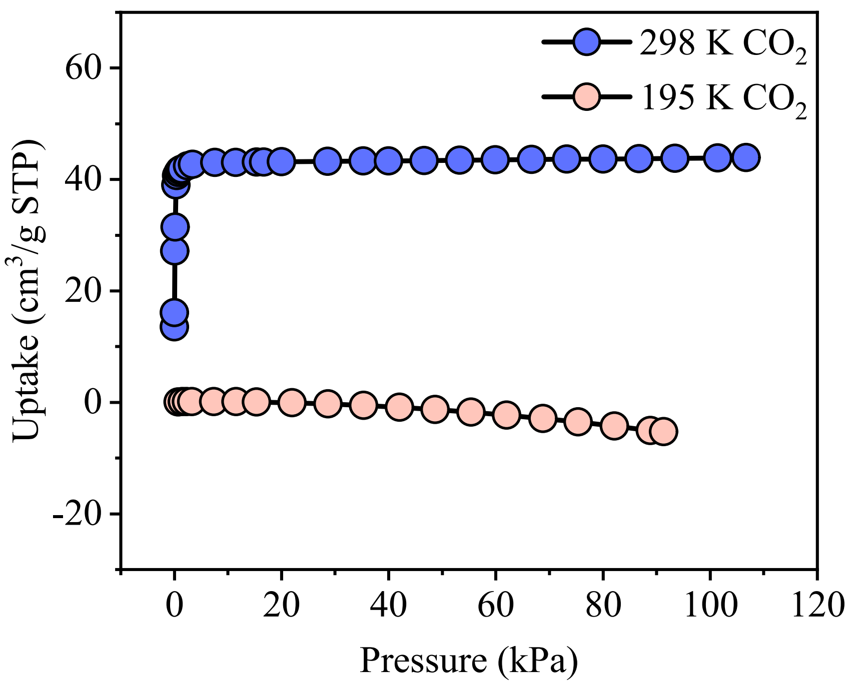


Figure S9. Single-component adsorption isotherms of CO_2 for **UTSA-400** at 195 K and 298 K.

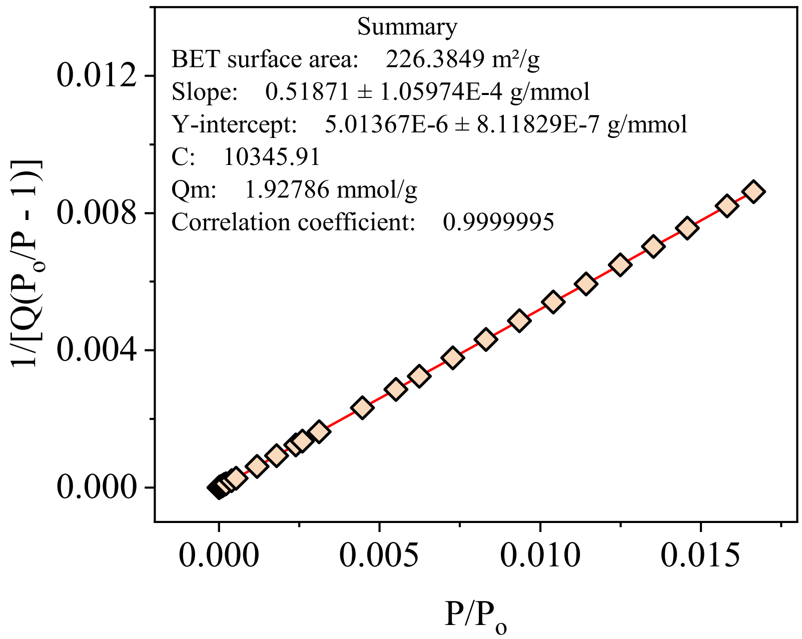


Figure S10. Calculation of BET surface area for **UTSA-400** based on CO_2 adsorption isotherm at 298 K.

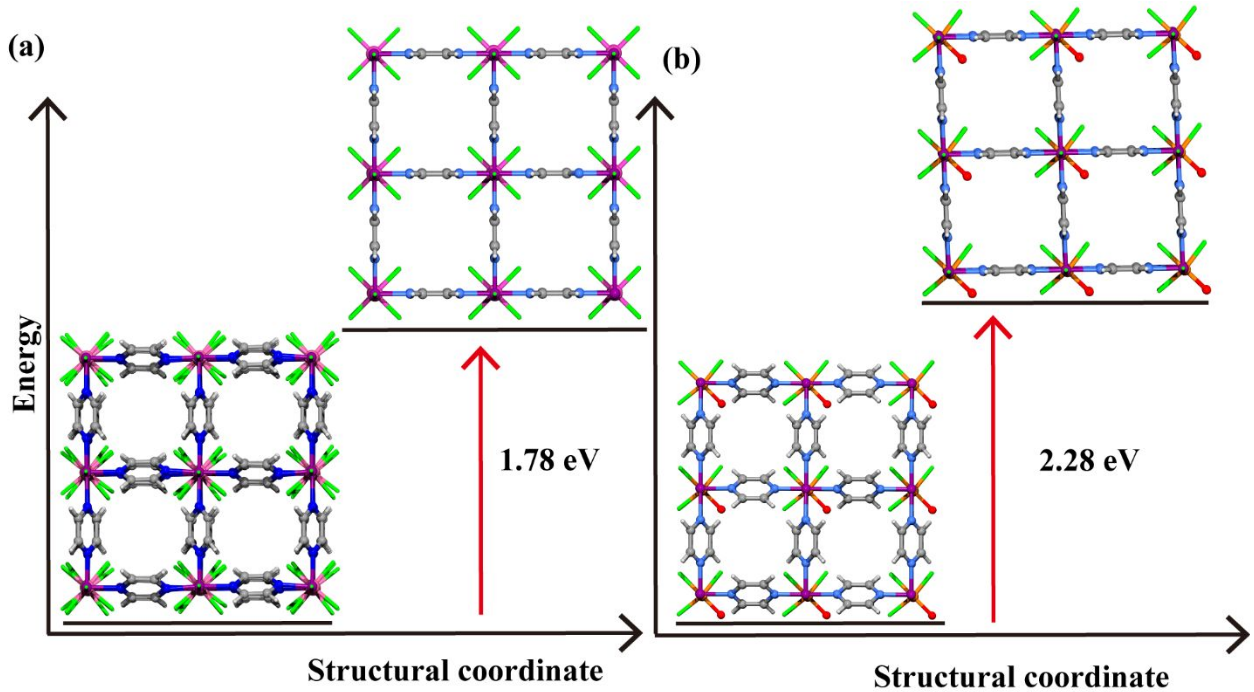


Figure S11. The energy difference between as-synthesized phase and open phase (tilting angle $\Theta = 0°$) for (a) **NbOFFIVE-1-Ni** and (b) **UTSA-400**.

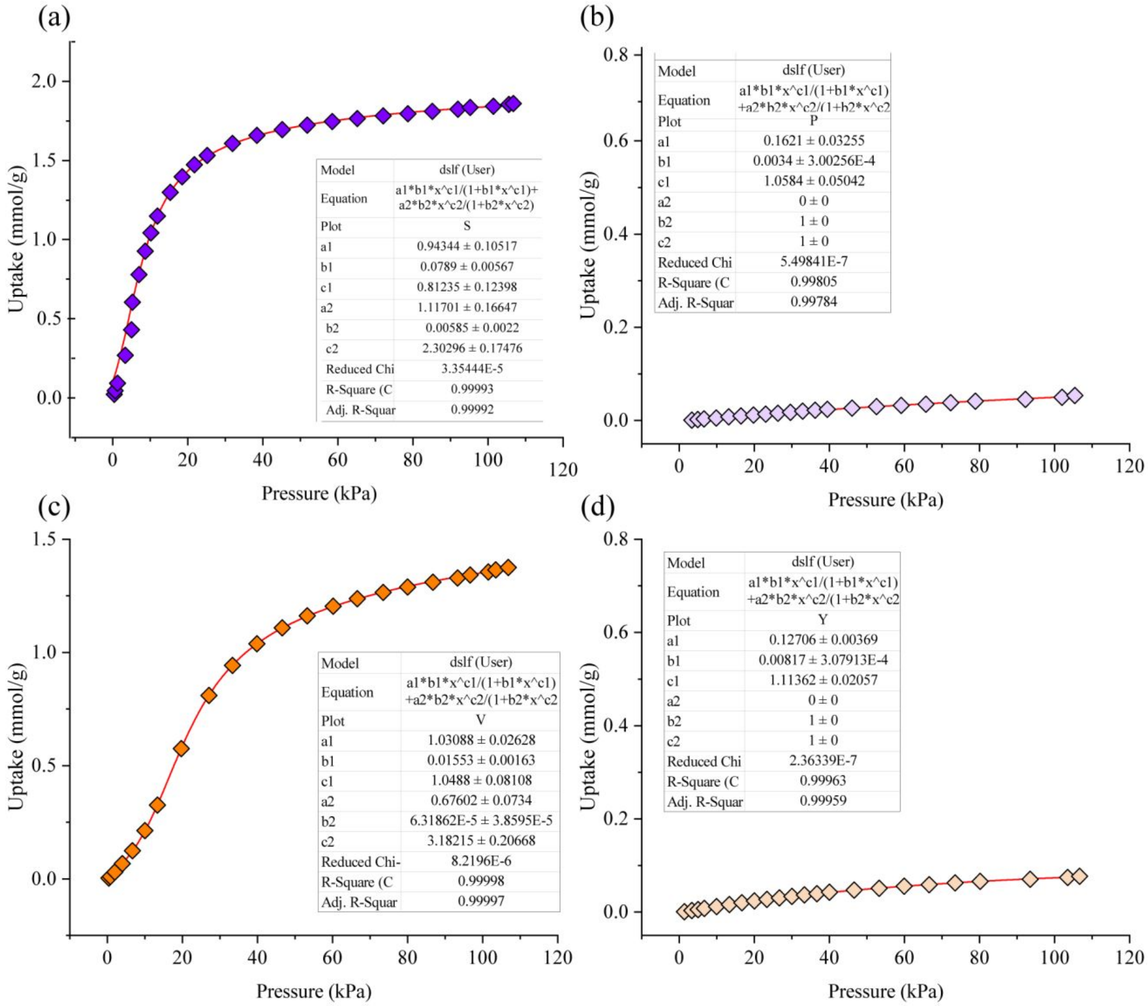


Figure S12. DSLF fitting of the C_3H_6 (a) and C_3H_8 (b) sorption data at 298 K and 1 bar for **UTSA-**400. DSLF fitting of the C_3H_6 (c) and C_3H_8 (d) sorption data at 298 K and 1 bar for **NbOFFIVE-1-Ni**.

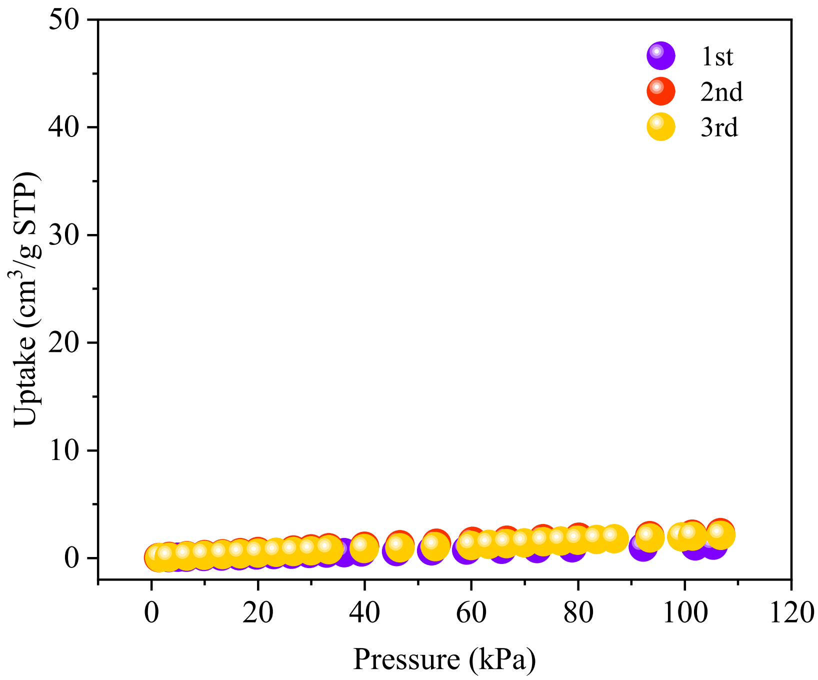


Figure S13. Multiple cycles of C_3H_8 adsorption isotherms for **UTSA-400** at 298 K.

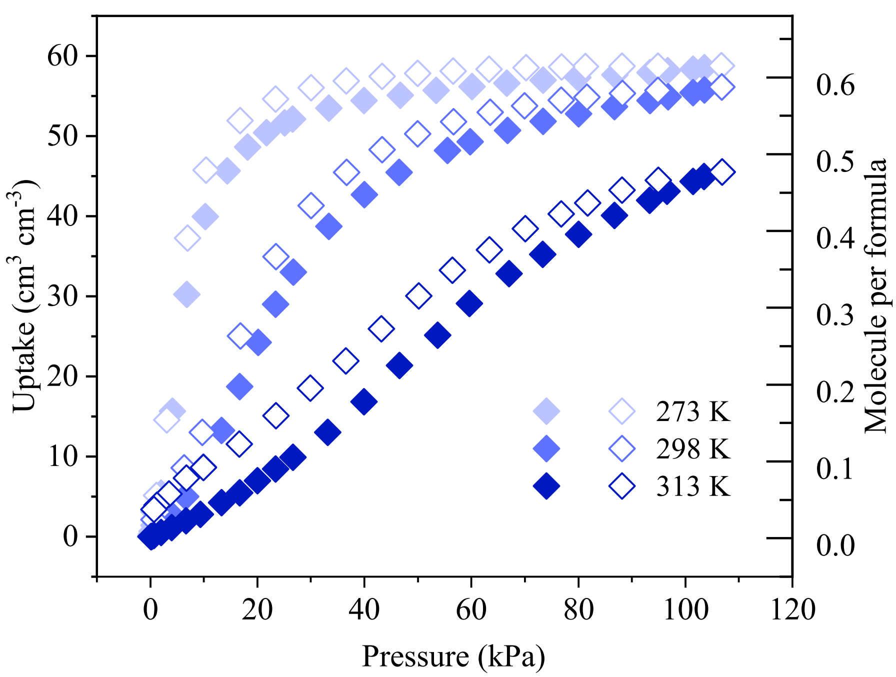


Figure S14. Single component adsorption isotherms of C_3H_6 at 273-313 K for **NbOFFIVE-1-Ni** with corresponding adsorption amount per pocket indicated.

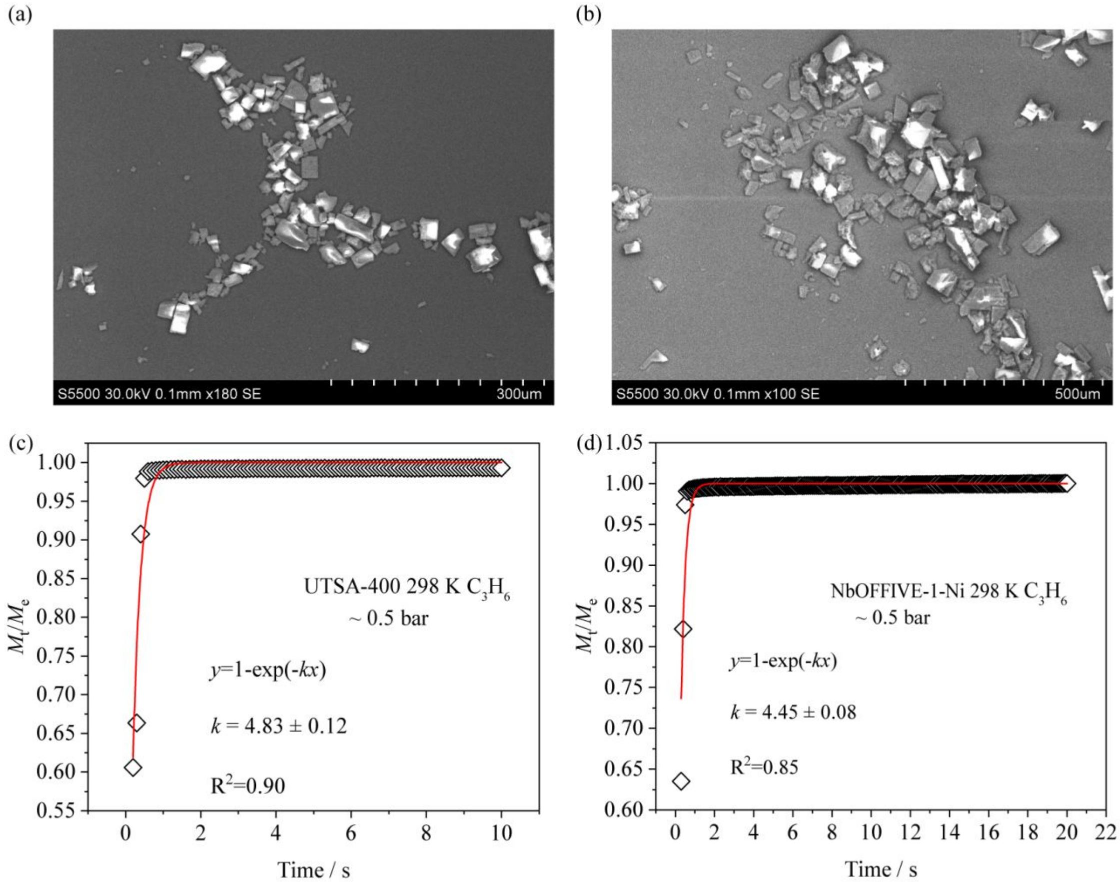


Figure S15. SEM image of **UTSA-400** (a) and **NbOFFIVE-1-Ni** (b). SEM measurement was performed on a Hitachi S5500 Field-emission Scanning Electron Microscope. Kinetic profile of **UTSA-400** (c) and **NbOFFIVE-1-Ni** (d) for C_3H_6 adsorption at 298 K and equilibrium pressure of ~0.5 bar. The diffusion coefficient was estimated to be 5.08×10^{-9} and 4.32×10^{-9} cm^2/s, respectively, based on linear driving force model. The diffusion coefficients are higher than one of the benchmark materials, Co-gallate, (*J. Am. Chem. Soc.* **2020**, 142, 17795) and comparable to other microporous MOFs (*J. Phys. Chem. Lett.* **2012**, 3, 2130).

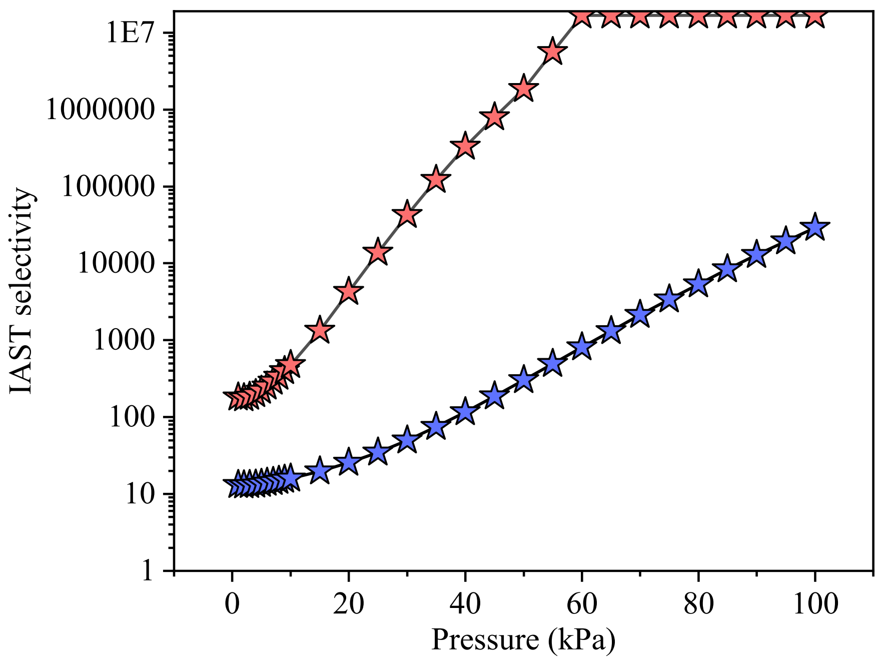


Figure S16. IAST selectivity of 50/50 C_3H_6/C_3H_8 mixture for **UTSA-400 (red)** and **NbOFFIVE-1-Ni** (blue) at 298 K.

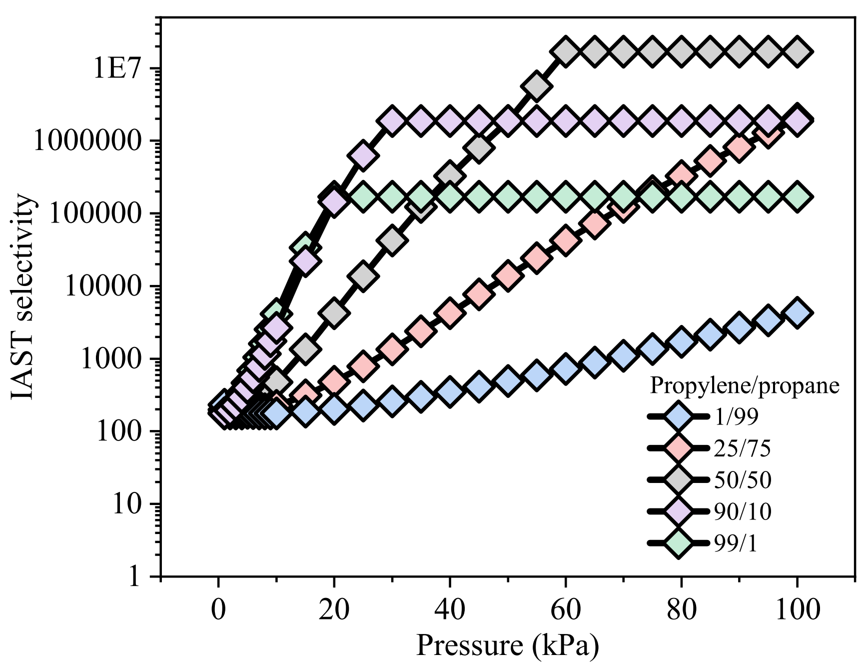


Figure S17. IAST selectivity of different C_3H_6/C_3H_8 mixtures for **UTSA-400** at 298 K.

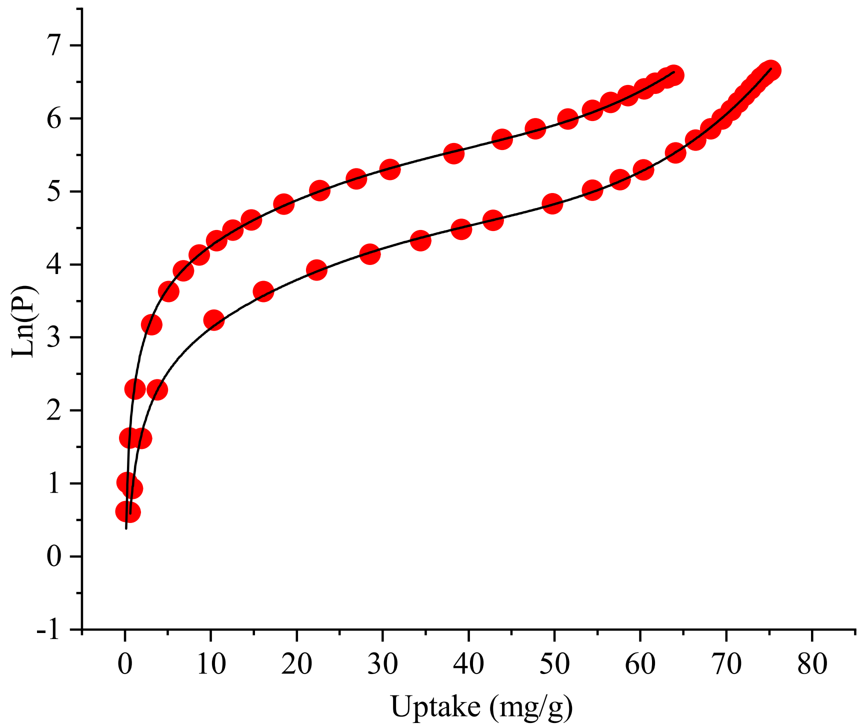


Figure S18. Virial fitting of the C_3H_6 isotherms C_3H_6 mixture for **UTSA-400** at 298 and 313 K.

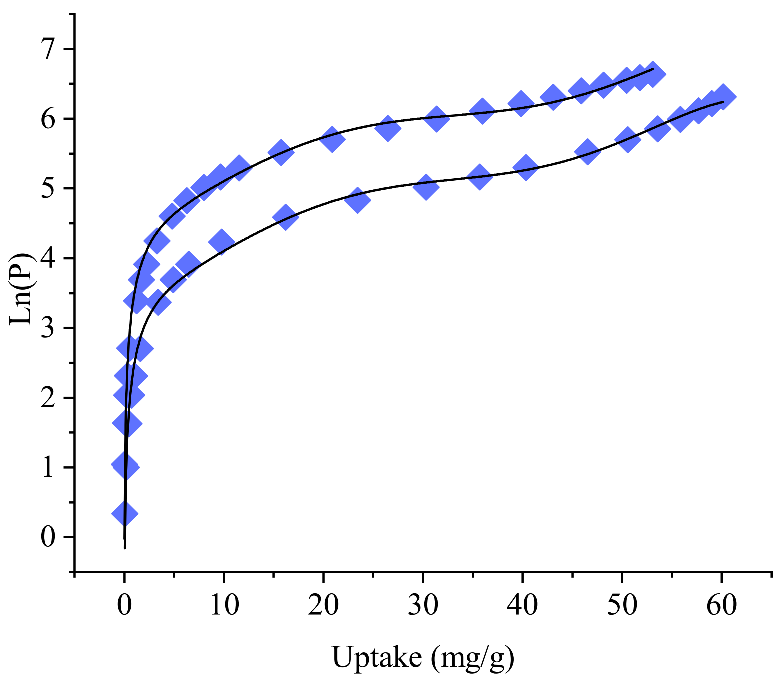


Figure S19. Virial fitting of the C_3H_6 isotherms C_3H_6 mixture for **NbOFFIVE-1-Ni** at 298 and 313 K.

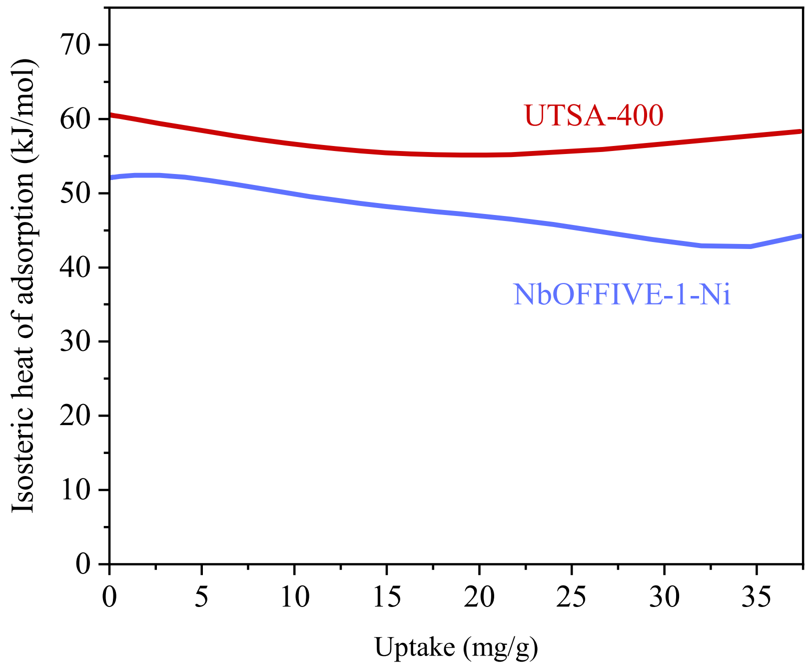


Figure S20. Isosteric heat of adsorption (Q_{st}) of C_3H_6 for **UTSA-400** (red) and **NbOFFIVE-1-Ni** (blue).

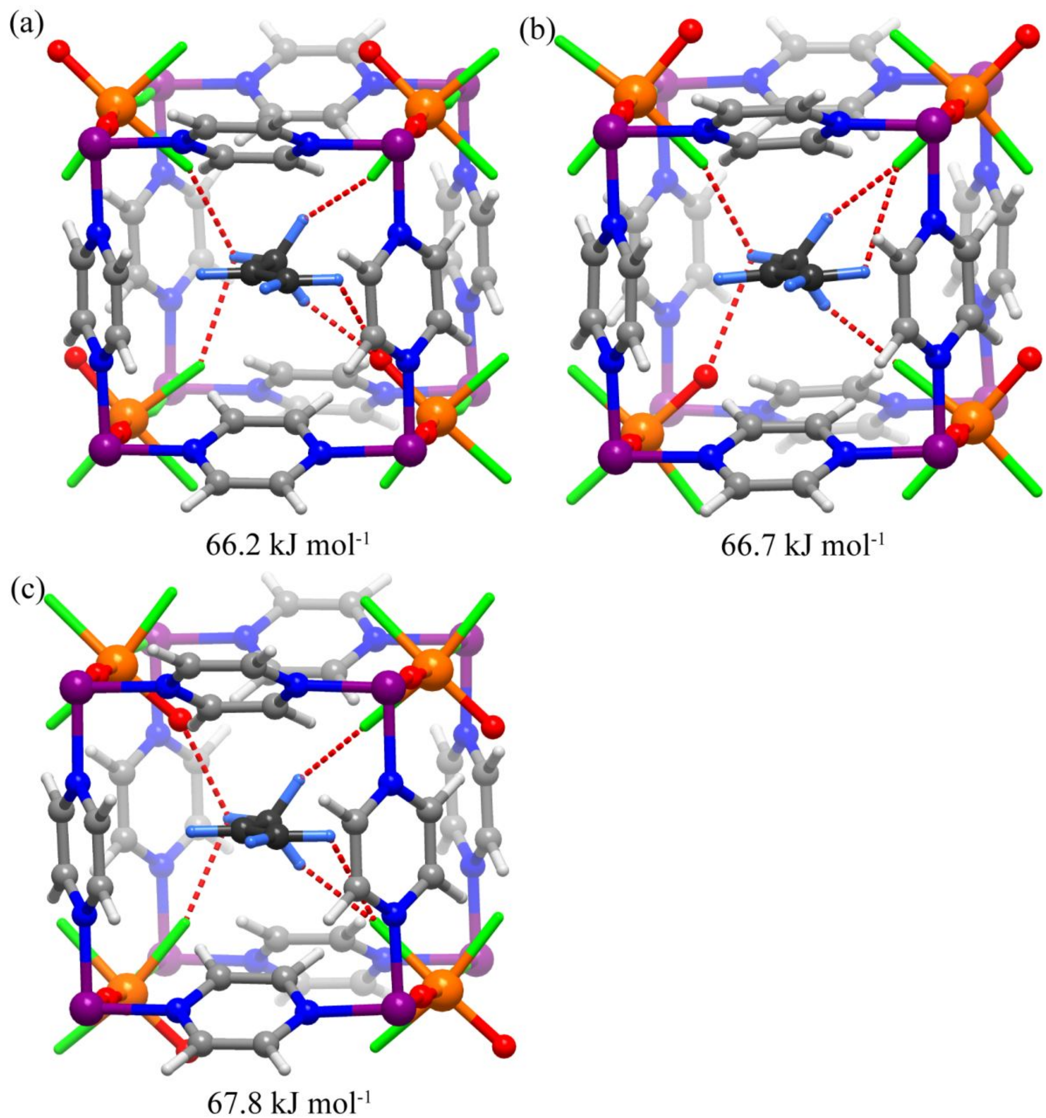


Figure S21. Three different propylene binding orientations with respect to the position of oxide in **UTSA-400** obtained by DFT-D calculations.

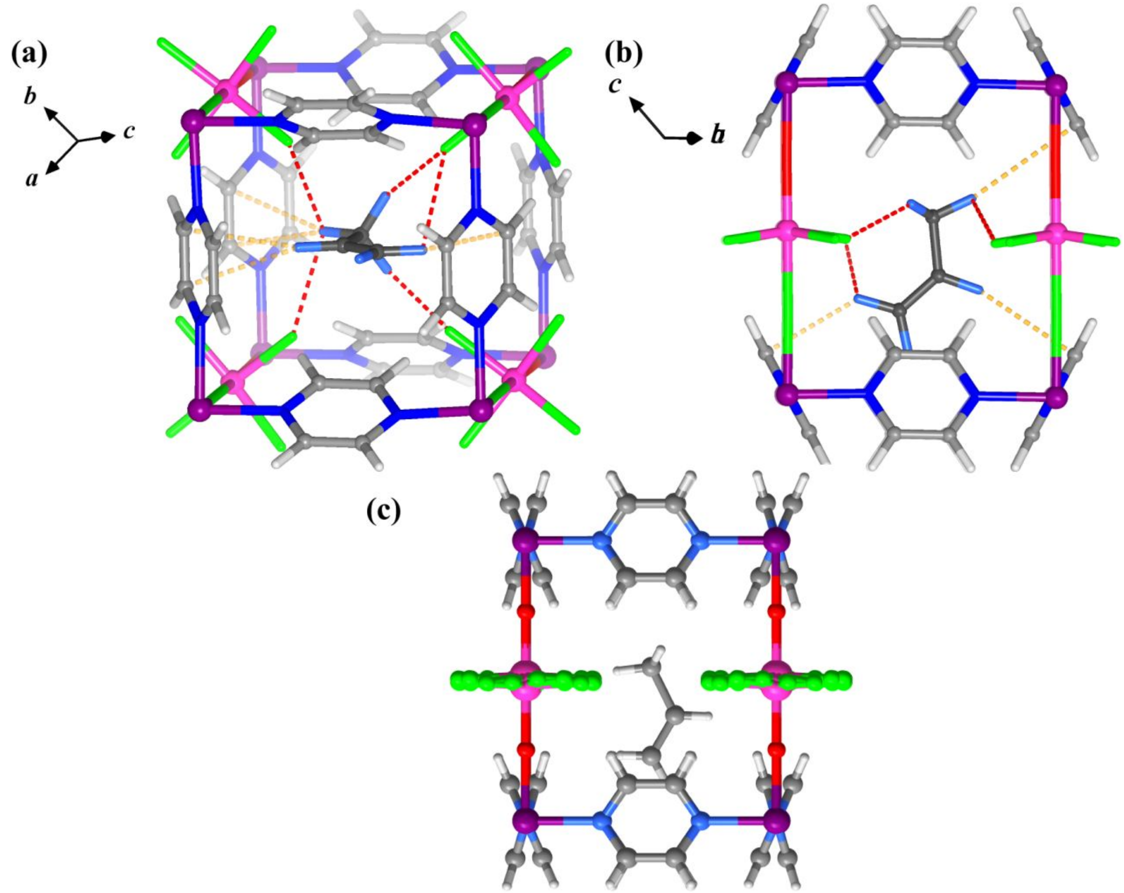


Figure S22. Perspective view (a) and top view (b) of DFT-D calculated binding site of C_3H_6 in **NbOFFIVE-1-Ni.** (c) Top view of C_3H_6@**NbOFFIVE-1-Ni** crystal structure reported in Ref 61.

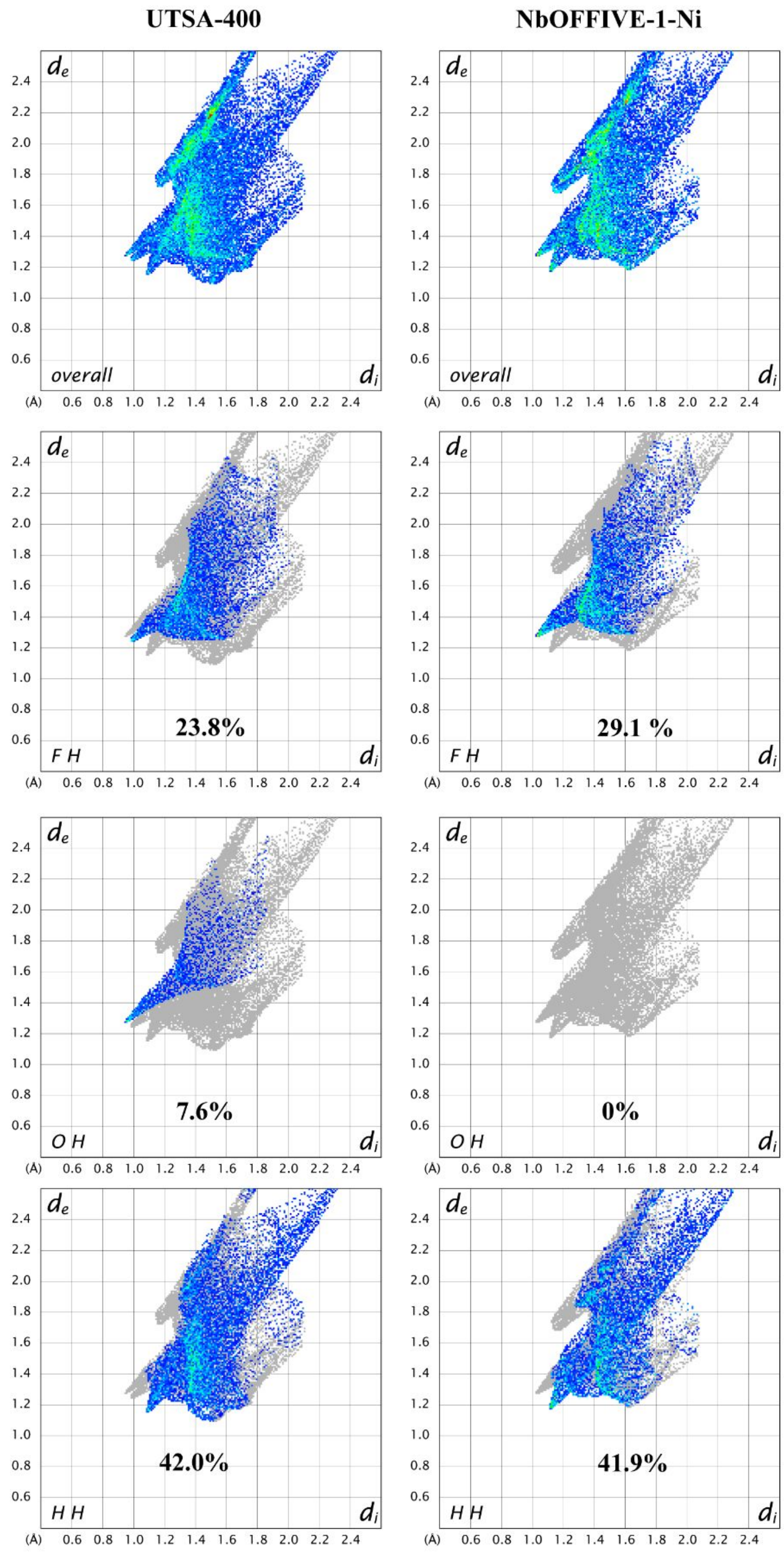


Figure S23. Derived 2D fingerprint plot of host-guest interactions in **UTSA-400** (left) and **NbOFFIVE-1-Ni** (right).

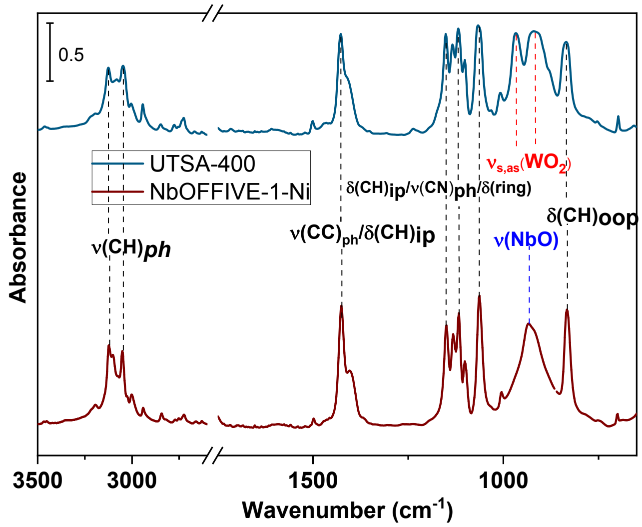


Figure S24. IR spectra of activated **UTSA-400** (top) and **NbOFFIVE-1-Ni** (bottom) for comparison. Notation and acronym: ν = stretching, δ = deformation, ip = in plane, and oop = out of plane, as = asymmetric, s = symmetric, and ph=phenyl.

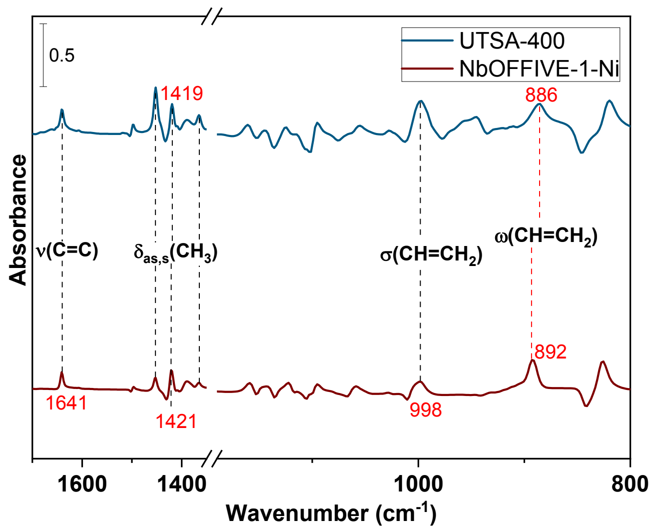


Figure S25. IR spectra of the adsorbed propene in **UTSA-400** (top) and **NbOFFIVE-1-Ni** (bottom) for comparison. The gas phase spectrum of propene at 760 ~Torr was subtracted.

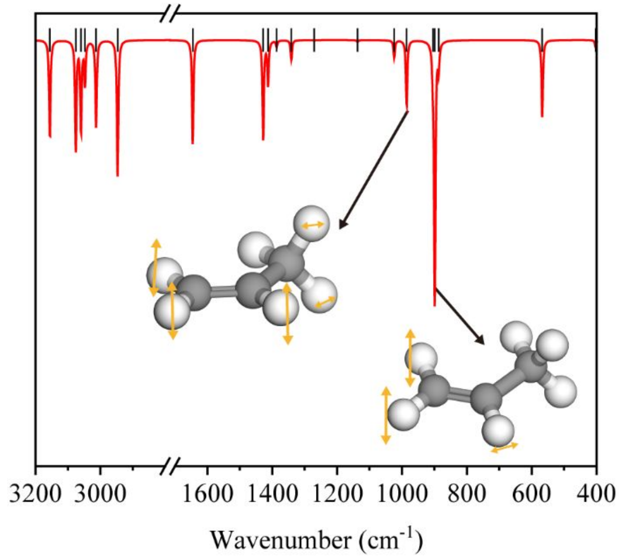


Figure S26. Simulated IR spectrum of propylene molecule with σ(CH=CH) (left) and ω(CH=CH) (right) vibration mode shown.

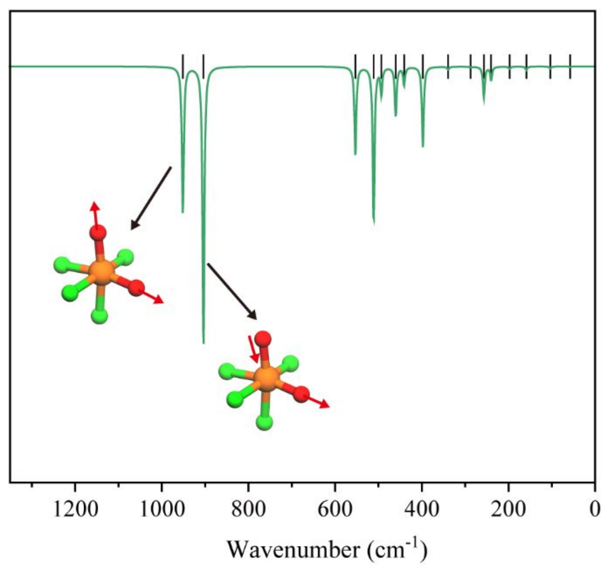


Figure S27. Simulated IR spectrum of $WO_2F_4^{2-}$ anion with symmetric and asymmetric vibration mode shown.

Table S4. Summary of selected **UTSA-400** and **NbOFFIVE-1-Ni** vibrational modes.

Assignment	Position	
	UTSA-400	**NbOFFIVE-1-Ni**
ν $(CH)_{ph}$	3125	3122
	3047	3050
	2942	3001
ν(CC)/δ$(CH)_{ip}$	1427	1425
δ$(CH)_{ip}$/ν$(CN)_{ph}$/δ(ring) The coupled vibrations	1151	1149
	1133	1131
	1118	1117
	1101	1100
	1066	1063
δ$(CH)_{oop}$	835	832
ν(WO_2)	967	-
	920	-
ν(NbO)	-	933
δ$(CH)_{oop}$	835	832

Table S5. Summary of vibrational bands of adsorbed propene in **UTSA-400** and **NbOFFIVE-1-Ni**.

Assignment	Observed frequency position (cm^{-1})	
	UTSA-400	**NbOFFIVE-1-Ni**
ν(CH/CH_2)	Region obscured	
ν$(CH_3)_{as,s}$		
ν(C=C)	1641	1641
$\delta_{as}(CH_3)$	1453	1453
$\delta_{as}(CH_3)$	1419	1421
$\delta_{s}(CH_3)$	1365	1365
σ(CH=CH)	998	998
ω(CH=CH)	886	892

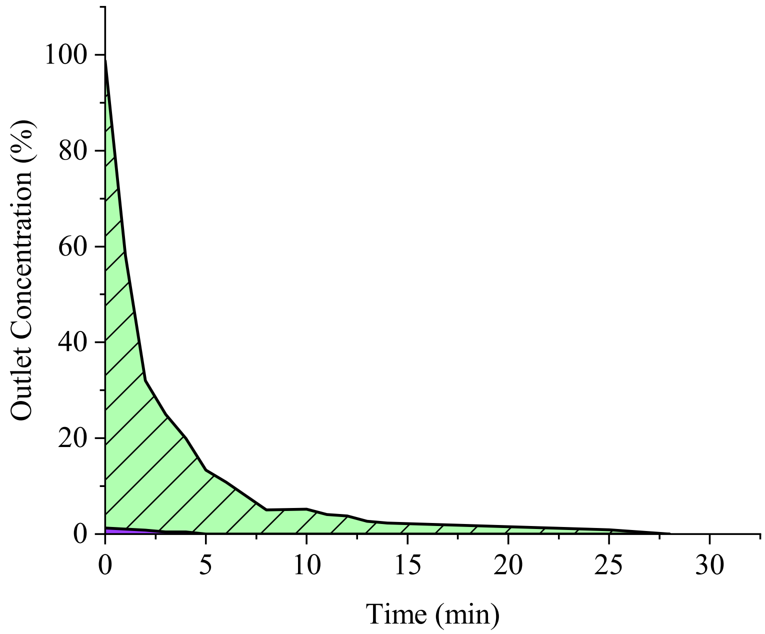


Figure S28. Concentration curve of the desorbed C_3H_6 (green) from **NbOFFIVE-1-Ni** during the regeneration process. Concentration curve of the desorbed C_3H_8 is shown in purple.

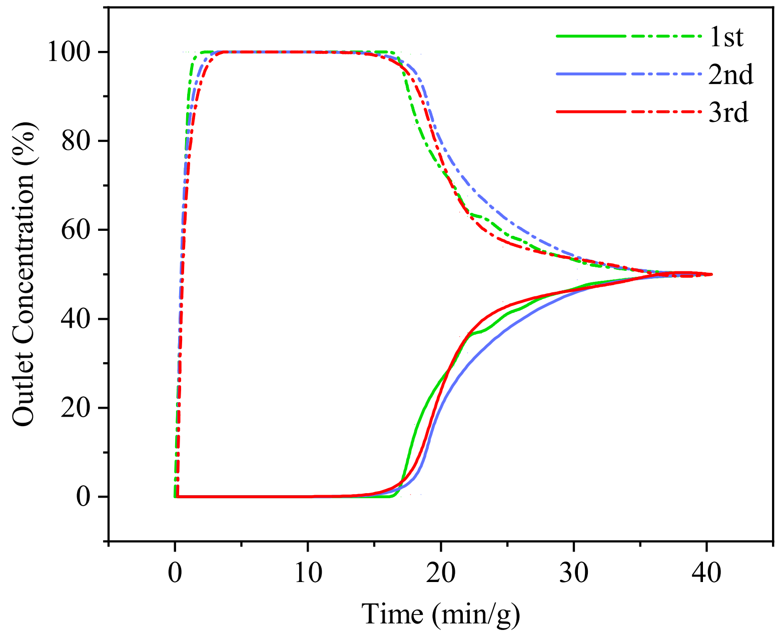


Figure S29. Cycling breakthrough experiments for **UTSA-400** for an equimolar binary mixture of C_3H_6 (solid line)/C_3H_8 (dash line) at 298 K and 1 bar after combined vacuum and temperature swing regeneration.

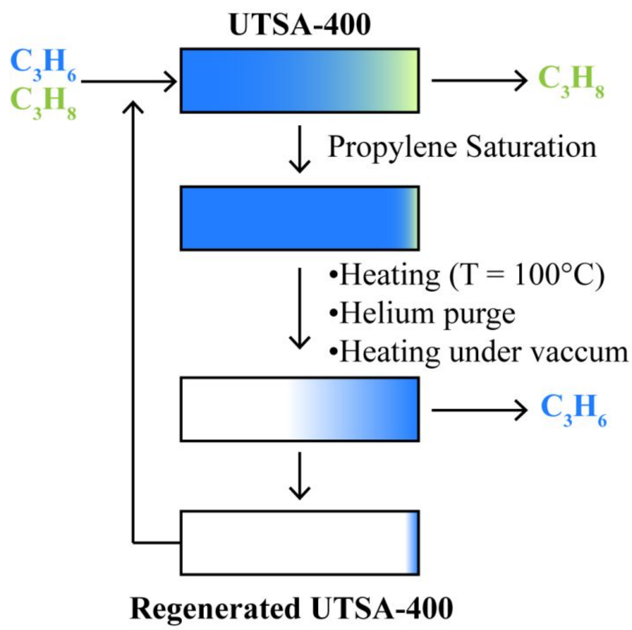


Figure S30. Schematic diagram of regeneration process for propylene separation using **UTSA-400** as adsorbent. Typical temperature swing adsorption (TSA) process (heating at 100 °C), helium purge, or combined vacuum and temperature swing adsorption (VTSA) process can be applied to regenerate the column.

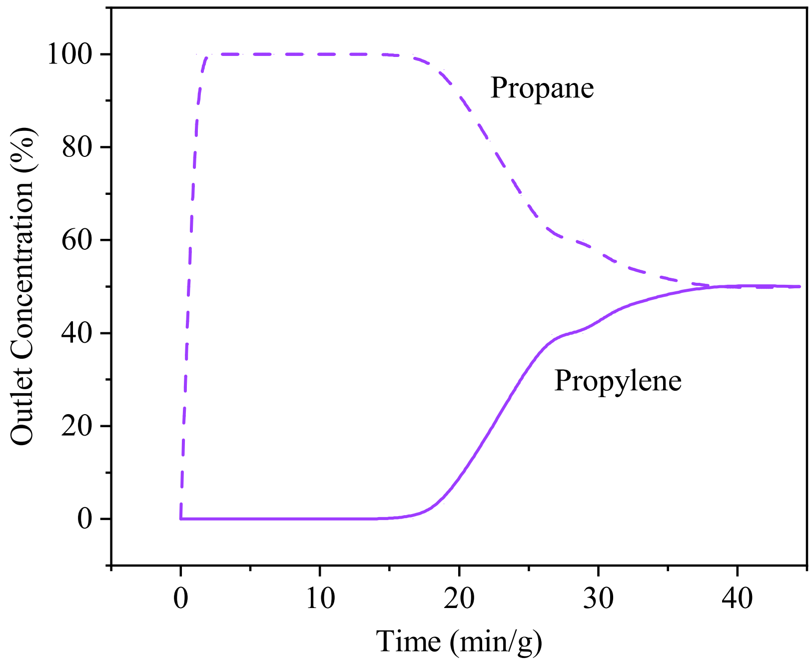


Figure S31. Breakthrough experiment for **UTSA-400** for an equimolar binary mixture of C_3H_6 (solid line)/C_3H_8 (dash line) at 298 K and 1 bar after temperature swing regeneration.

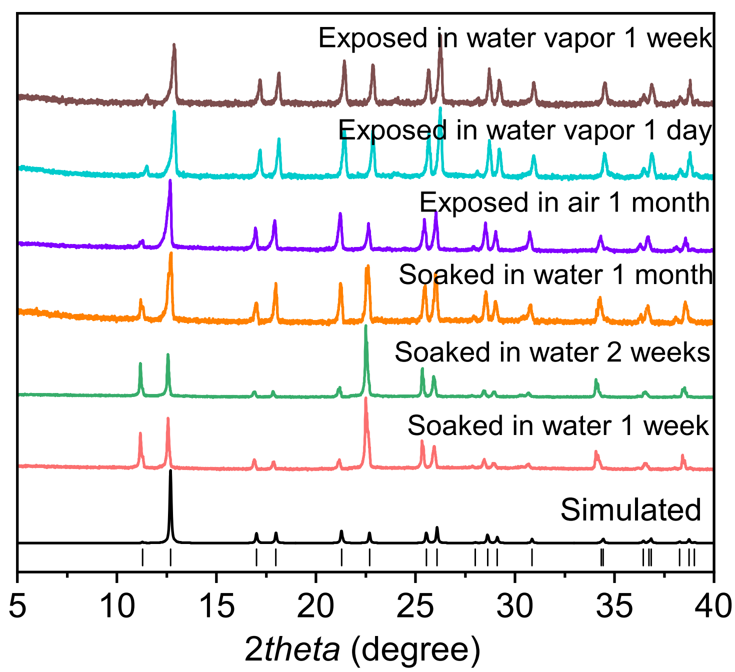


Figure S32. PXRD patterns of **UTSA-400** after water stability tests.

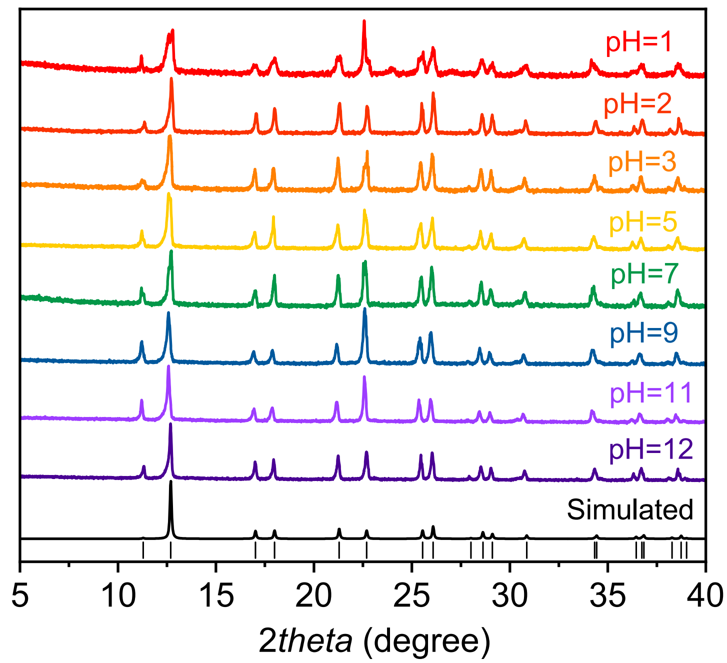


Figure S33. PXRD patterns of **UTSA-400** after chemical stability tests (soaked for 24h).

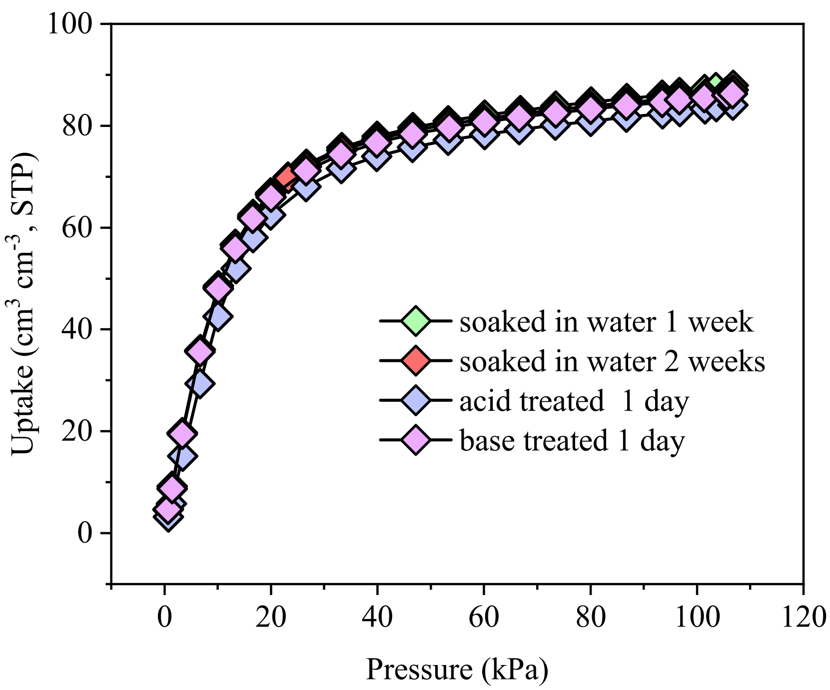


Figure S34. C_3H_6 adsorption isotherms for **UTSA-400** after water stability tests and treated with acid (pH =1)/base (pH = 12) for 1 day.

Table S6. Summary of the gas uptakes, selectivities and Q_{st} for C_3H_6 in various reported MOFs.

Materials	*T* (K)& *P* (bar)	C_3H_6 uptake ($cm^3\ cm^{-3}$)	C_3H_8 uptake ($cm^3\ cm^{-3}$)	IAST selectivity[a] at 1 bar	Q_{st} for C_3H_6 ($kJ\ mol^{-1}$)
MAF-23-O[9]	298, 1	44.1	32.7	8.8	54
ZnAtzPO_4[10]	298, 1	75.5	42.2	--	27.5
MFM-520[11]	298, 1	2.33[b]	2.03[b]	17	48.5
Zeolite					
zeolite 13X[12]	303, 1	82.1	65.8	--	42.5
zeolite 4A[12]	303, 1	73.2	~3.8	--	29.9
zeolite 5A pellet[13]	323, 1	96.7	91.2	--	~49
zeolite 5A crystal[13]	323, 1	128.4	112.5	--	~49
Si-CHA[14]	303, 0.86	90.3	--	--	33.5
ITQ-3[14]	303, 0.86	51.9	--	--	28.5
OMS-type MOFs					
Ni-NP	298, 1	3.57[b]	2.13[b]	10.5	57
Fe_2(dobdc)[15]	318, 1	174	153	15	44
Co_2(dobdc)[16]	298, 1	192	141	45	--
Mn_2(*m*-dobdc)[17]	298, 1	188	154	43	67.4
Fe_2(*m*-dobdc)[17]	298, 1	190	154	60	73.0
Co_2(*m*-dobdc)[17]	298, 1	200	160	43	53.1
Ni_2(*m*-dobdc)[17]	298, 1	197	163	38	55.1
MIL-100 (Fe)[18]	303, 0.065	1.37[b]	0.63[b]	--	70
AGTU-3[19]	298, 1	41.9	15.8	7	68
$Cu_3(btc)_2$[20]	323, 1	124	98.4	--	41.8
Flexible MOFs					
CPL-1[21]	273, 1	73.2	11.7	--	22.8
NKU-FlexMOF-1[22]	298, 1	~94	~81	--	61.6
JNU-3a[23]	303, 1	78.6	64.8	513	29.3[2]

Molecular Sieving type MOFs					
UTSA-400	**298, 1**	**92.1**	**2.5**	**$>10^7$**	**60.5**
Y-dbai[24]	298, 1	75.9	~3	$>10^{10}$	42.4
Y-abtc[25]	298, 1	64.6	--	--	50.0
Co-gallate[26]	298, 1	66.6	5.2	333[a]	41.0
NbOFFIVE-1-Ni	298, 1	54.3	3.0	$>10^4$	52.0
HIAM-301[27]	298,1	93.0	<8.8	>150	27

[a]IAST selectivities are only for the qualitative comparison purpose; [b]The unit for uptake is mmol g^{-1}.

Table S7. Summary of virial fitting parameters of **UTSA-400** and **NbOFFIVE-1-Ni**.

	UTSA-400	**NbOFFIVE-1-Ni**
a_0	-7283.6112	-6263.2642
a_1	26.3996	-27.4409
a_2	0.27101	4.6543
a_3	-0.0211	-0.1794
a_4	2.2202E-4	0.00297
a_5	-5.9585E-7	-1.7684E-5

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